2 Cell MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it and fill in the corresponding circle on the answer sheet.

Dialysis tubing is permeable to water molecules but not to sucrose. Four dialysis tubes are half filled with 5 percent, 10 percent, 20 percent, and 40 percent sucrose solutions, respectively, and two dialysis tubes are half filled with distilled water. The dialysis tubes are all sealed at both ends, and the initial masses are determined. Five dialysis tubes are placed into beakers containing distilled water, and the sixth dialysis tube, containing distilled water, is placed into a 40 percent sucrose solution. The masses of the dialysis tubes are recorded at 30-minute intervals for 90 minutes, as shown in the table below.

Dialysis Tube Number	Dialysis Tube Contents	Beaker Contents	Initial Mass	Mass (grams)		
				30 Min	60 Min	90 Min
1	5% Sucrose	Distilled water	12.8	13.3	13.4	13.8
2	10% Sucrose	Distilled water	15.6	16.7	17.0	17.3
3	20% Sucrose	Distilled water	13.7	15.4	16.2	16.5
4	40% Sucrose	Distilled water	13.4	16.3	18.1	19.6
5	Distilled water	Distilled water	11.2	11.2	11.2	11.2
6	Distilled water	40% Sucrose	13.8	10.1	8.3	6.3

1. The contents of which dialysis tube are initially isotonic to the distilled water in the beaker?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5 ✓

2. A net movement of water into the beaker occurs in which of the following dialysis tubes?

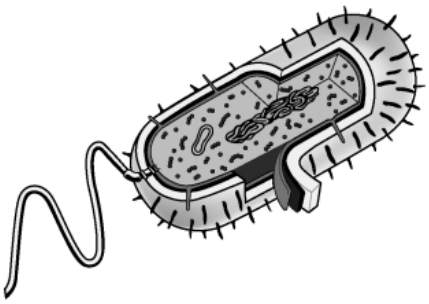
2 Cell MCQ

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6 ✓

3. To model a plant cell, a permeable, nonflexible case is placed around each piece of dialysis tubing. The greatest pressure potential will develop within which dialysis tube number?

(A) 2
(B) 3
(C) 4 ✓
(D) 5
(E) 6

4.

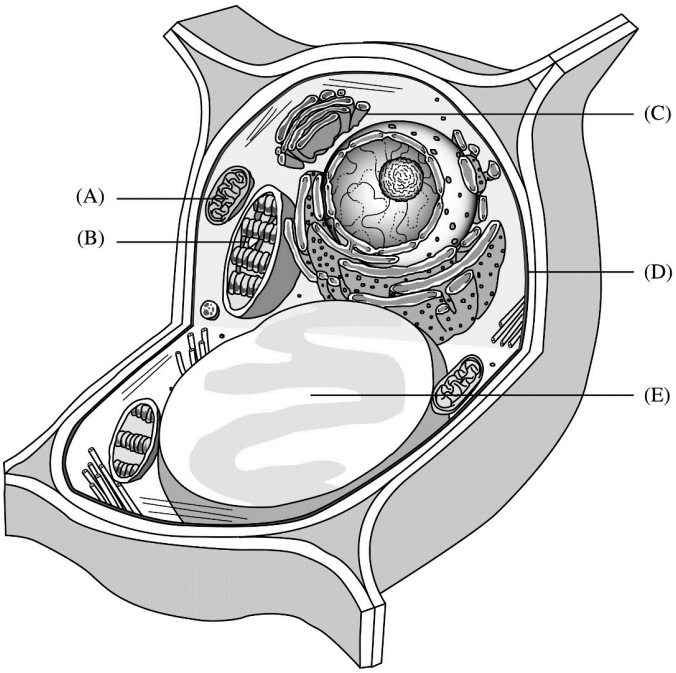


Which of the following is true about the structure of the prokaryotic cell represented in the figure?

(A) It possesses cilia and lysosomes.
(B) It has a diploid genome and contains linear DNA.
(C) It has a cell wall exterior to the plasma membrane and contains ribosomes. ✓
(D) It possesses membrane-bound organelles and a flagellum.

2 Cell MCQ

The following questions refer to the following diagram. For each phrase or sentence, select the labeled part to which it is most closely related. Each option may be used once, more than once, or not at all for each group.



5. Evolved from a photoautotrophic prokaryote

(A) A

(B) B ✓

(C) C

(D) D

(E) E

6. Site of modification and packaging of proteins and lipids prior to export from the cell

(A) A

(B) B

(C) C ✓

(D) D

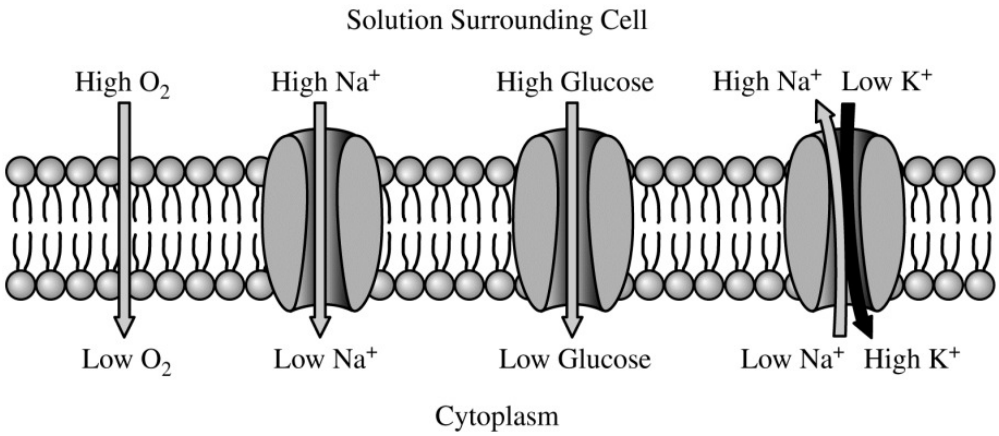
(E) E

7. Site of transport of materials into and out of the cell

2 Cell MCQ

(A) A
(B) B
(C) C
(D) D ✓
(E) E

8.



The manner in which several different ions and molecules move through a cell membrane is shown in the diagram above. For each ion or molecule, the relative concentration on each side of the membrane is indicated. Which of the following accurately describes one of the movements taking place?

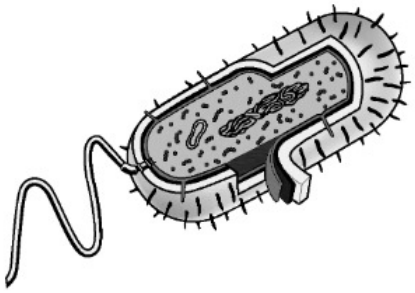
(A) Glucose is transported into the cell by active transport.
(B) Na^+ is transported into the cell by active transport.
(C) The movement of glucose through the membrane requires ATP hydrolysis.
(D) Na^+ transport out of the cell requires ATP hydrolysis. ✓

9. Intravenous (IV) solutions administered to patients are normally isotonic. Which of the following is most likely if an IV of distilled water is administered to a patient?

(A) The cells that are exposed to hypotonic solutions will shrink as a result of salt moving into the blood.
(B) The liver will secrete additional bile salts into the blood to raise the tonicity of the administered fluid.
(C) The cells that are exposed to hypotonic solutions will expand as water moves osmotically into the cells from the blood. ✓
(D) The patient's respiration rate will slow to compensate for the higher levels of circulating blood.

2 Cell MCQ

10.



The diagram above represents a typical rod-shaped bacterium. Which of the following best describes a feature shown in the diagram that is unique to archaea and bacteria?

(A) The organism is surrounded by a cell wall.

(B) The organism contains ribosomes.

(C) The organism does not have a nuclear membrane surrounding its genetic material. ✓

(D) The organism is not capable of making or providing itself with ATP.

11. Water crosses the plasma membranes of certain eye cells at a higher rate than it does in other cell types. Which of the following best explains why the rate of water transport is higher in the eye cells?

(A) The eye cells move water against the concentration gradient, which is faster than moving water with the concentration gradient.

(B) The eye cells have protein channels that allow water to pass through in greater quantities than the phospholipid bilayer allows. ✓

(C) The eye cells have more ATP than other cells do, which allows them to use active transport to move water across the membrane more rapidly.

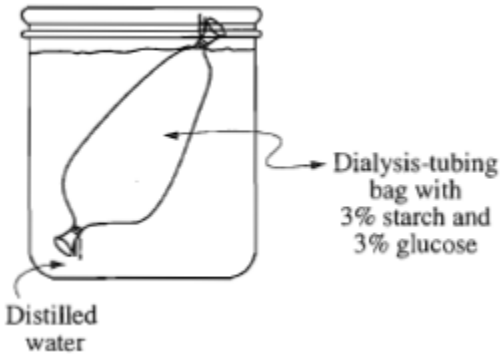
(D) The eye cells modify water molecules so that they are positively charged, allowing them to cross the phospholipid bilayer more easily than unmodified water molecules.

Answer B

Correct. Water is able to cross the cell membrane when moving with its concentration gradient. The presence of protein channels such as aquaporins allows water to move faster than when it simply passes between the phospholipids of the bilayer.

2 Cell MCQ

The following questions refer to an experiment in which a dialysis-tubing bag is filled with a mixture of 3% starch and 3% glucose and placed in a beaker of distilled water, as shown below. After 3 hours, glucose can be detected in the water outside the dialysis-tubing bag, but starch cannot.



12. From the initial conditions and results described, which of the following is a logical conclusion?

(A) The initial concentration of glucose in the bag is higher than the initial concentration of starch in the bag.

(B) The pores of the bag are larger than the glucose molecules but smaller than the starch molecules. ✓

(C) The bag is not selectively permeable.

(D) A net movement of water into the beaker has occurred.

(E) The molarity of the solution in the bag and the molarity of the solution in the surrounding beaker are the same.

13. Which of the following best predicts the condition expected after 24 hours?

(A) The bag will contain more water than it did in the original condition. ✓

(B) The contents of the bag will have the same osmotic concentration as the surrounding solution.

(C) Water potential in the bag will be greater than water potential in the surrounding solution.

(D) Starch molecules will continue to pass through the bag.

(E) A glucose test on the solution in the bag will be negative.

14. If, instead of the bag, a potato slice were placed in the beaker of distilled water, which of the following would be true of the potato slice?

(A) It would gain mass. ✓

(B) It would neither gain nor lose mass.

(C) It would absorb solutes from the surrounding liquid.

(D) It would lose water until water potential inside the cells is equal to zero.

(E) The cells of the potato would increase their metabolic activity.

2 Cell MCQ

15. A student used a microscope to observe a wet-mount slide of red onion epidermal cells that were suspended in a 1% NaCl solution. The student then added a 15% NaCl solution to the slide and observed the changes that occurred. The student's observations are represented in Figure 1.

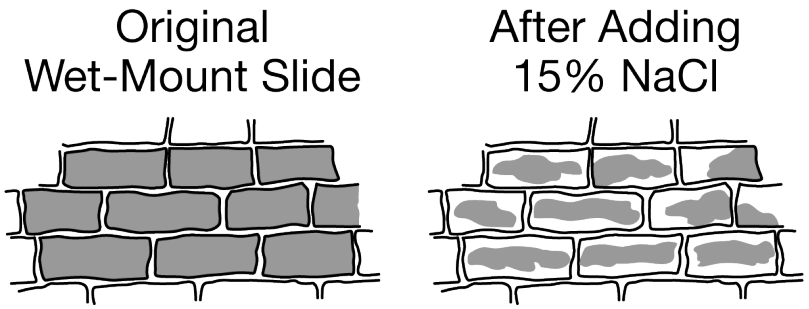


Figure 1. Student's observations of onion cells

Which of the following most directly explains the changes in the cells?

(A) The degradation of DNA in the nuclei of the cells

(B) The lysis of chloroplasts in the cells

(C) The movement of water from the central vacuoles of the cells into the solution ✓

(D) The movement of NaCl from the solution into the cytoplasm of the cells

16. Which of the following best supports the statement that mitochondria are descendants of endosymbiotic bacteria-like cells?

(A) Mitochondria and bacteria possess similar ribosomes and DNA. ✓

(B) Mitochondria and bacteria possess similar nuclei.

(C) Glycolysis occurs in both mitochondria and bacteria.

(D) Both mitochondria and bacteria have microtubules.

(E) Neither mitochondria nor bacteria possess chloroplasts.

17. Which of the following components of the cell membrane is responsible for active transport?

(A) Phospholipid

(B) Protein ✓

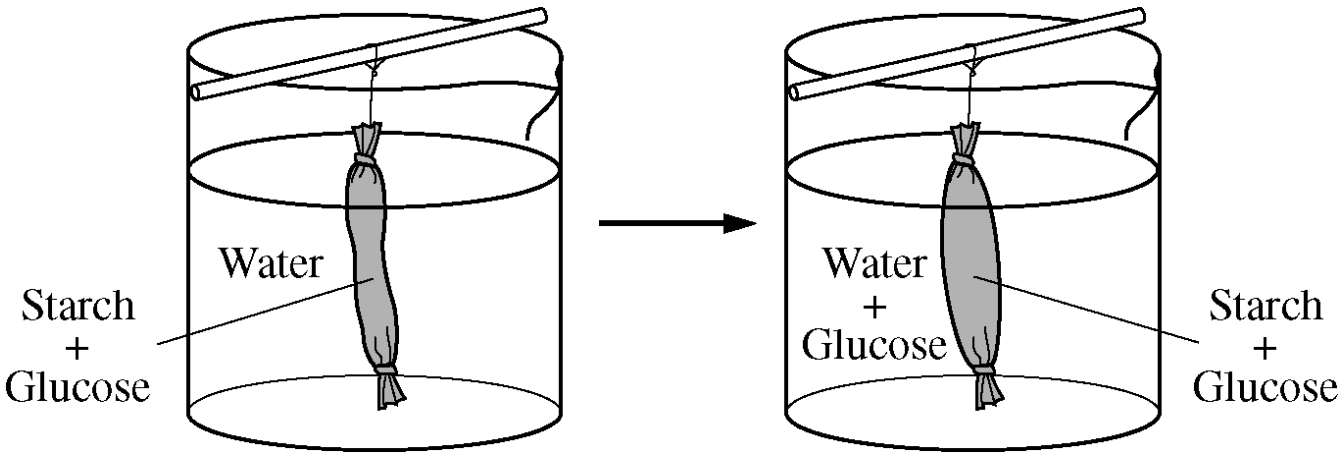
(C) Lipid

(D) Phosphate

(E) Cholesterol

2 Cell MCQ

18. A common laboratory investigation involves putting a solution of starch and glucose into a dialysis bag and suspending the bag in a beaker of water, as shown in the figure below.

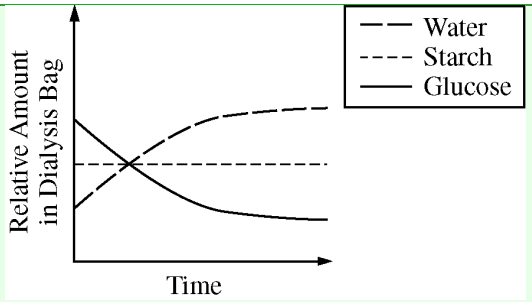


The investigation is aimed at understanding how molecular size affects movement through a membrane.

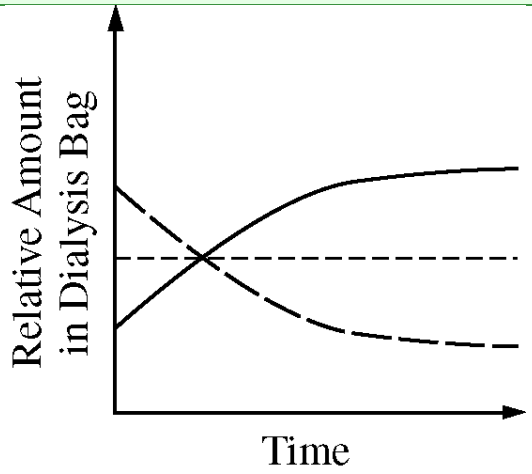
Which of the following best represents the amount of starch, water, and glucose in the dialysis bag over the course of the investigation?

2 Cell MCQ

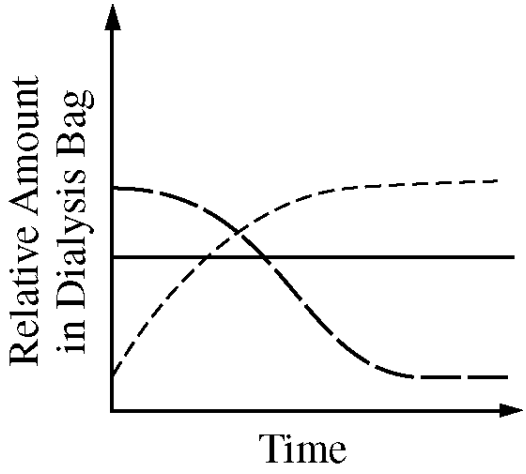
(A)



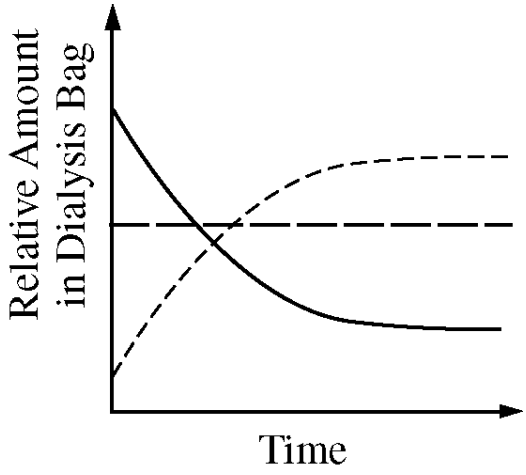
(B)



(C)

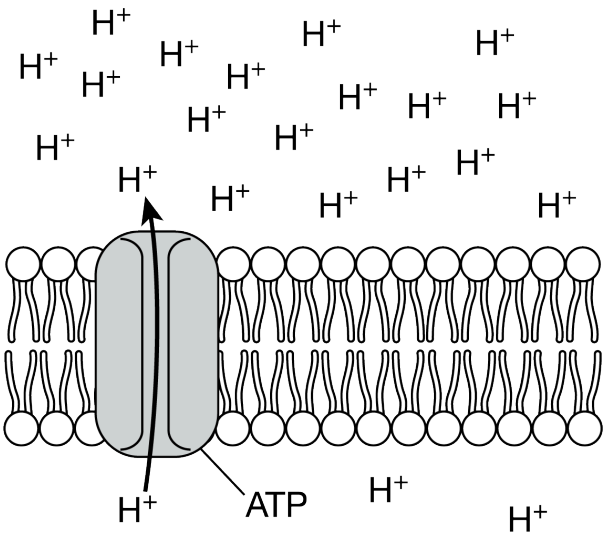


(D)



2 Cell MCQ

19. The illustration shows the active transport of hydrogen ions through a membrane protein.



Which of the following best predicts the effect of not having ATP available to supply energy to this process?

(A) H^+ ions will stop moving through the protein. ✓

(B) H^+ ions will move in the other direction through the protein.

(C) H^+ ions will continue to move through the protein in the original direction but at a slower rate.

(D) H^+ ions will begin to move through the phospholipid portion of the membrane in the original direction.

Answer A

Correct. Metabolic energy such as ATP is required for active transport of ions against a concentration gradient.

20. Which of the following transport mechanisms will be affected most directly by a temporary shortage of ATP molecules inside the cell?

(A) The movement of water molecules through aquaporins

(B) The diffusion of oxygen molecules across the plasma membrane

(C) The transport of glucose molecules against a concentration gradient ✓

(D) The facilitated diffusion of Ca^{2+} ions into the cell

2 Cell MCQ

Answer C

Correct. The active transport of glucose molecules against a concentration gradient requires an input of energy. Biological processes that require an input of energy typically incorporate the hydrolysis of ATP, an energy-rich molecule.

21. Some membrane proteins help maintain the concentrations of ions inside a cell by transporting the ions across the cell's plasma membrane. Other membrane proteins form pores in the plasma membrane through which the ions can diffuse. A model showing the influence of membrane proteins on the movement of sodium (Na^+) and potassium (K^+) ions across a plasma membrane is presented in Figure 1.

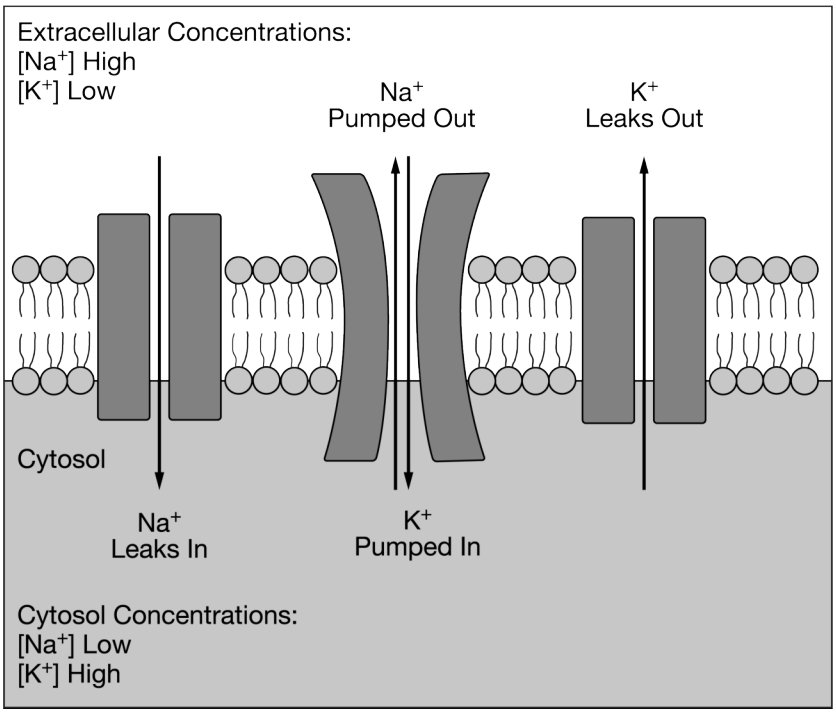


Figure 1. Section of a cell's plasma membrane, showing ion concentrations and membrane proteins

Based on the model presented in Figure 1, which of the following changes will most likely result from a depletion of available ATP stores inside the cell?

(A) The Na^+ concentration outside the cell will increase.

(B) The Na^+ concentration inside the cell will increase. ✓

(C) The K^+ concentration inside the cell will increase.

(D) The K^+ concentration outside the cell will decrease.

2 Cell MCQ

Answer B

Correct. Without ATP, the Na^+ ions will continue to leak into the cell but will not be pumped out of the cell, resulting in a concentration increase inside the cell.

22. The active transport pump used to move sodium ions across the membranes of gill cells in a freshwater fish has each of the following characteristics EXCEPT:

(A) It uses osmosis to carry sodium ions into the cells. ✓

(B) It requires energy.

(C) It has a specific binding site for ATP.

(D) It has a specific binding site for sodium ions.

(E) It is a protein.

2 Cell MCQ

23. Figure 1 shows a model of how a channel protein influences the movement of a particle across a cell's plasma membrane.

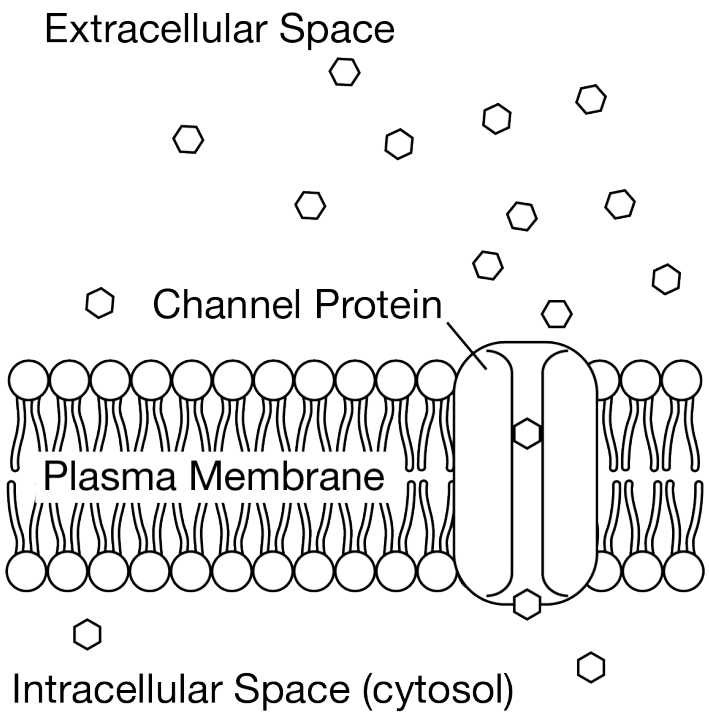


Figure 1. A section of a cell's plasma membrane, showing a channel protein and a concentration gradient across the membrane

An investigator wants to understand whether a newly found membrane protein is involved in membrane transport of a certain particle. Which investigation will help determine whether the new membrane protein is a channel protein involved in membrane transport?

(A) Add small nonpolar molecules to the extracellular space and measure the direction of particle movement of the molecules.

(B) Measure the rate of extracellular fluid movement into the intracellular space.

(C) Add more of the proteins to the plasma membrane and measure the rate of the particle movement. ✓

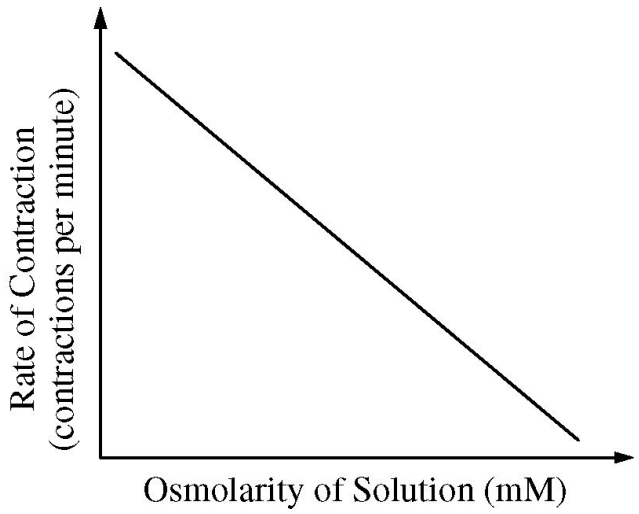
(D) Remove ATP from the intracellular space and measure the rate of the particle movement into the intracellular space.

Answer C

Correct. If the movement of particles is mediated by the channel protein, then an increase in proteins in the cell membrane should influence the rate of particle movement across the membrane.

2 Cell MCQ

24.



Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experimental investigation, paramecia were placed in salt solutions of increasing osmolarity. The rate at which the contractile vacuole contracted to pump out excess water was determined and plotted against osmolarity of the solutions, as shown in the graph. Which of the following is the correct explanation for the data?

(A) At higher osmolarity, lower rates of contraction are required because more salt diffuses into the paramecia.

(B) The contraction rate increases as the osmolarity decreases because the amount of water entering the paramecia by osmosis increases. ✓

(C) The contractile vacuole is less efficient in solutions of high osmolarity because of the reduced amount of ATP produced from cellular respiration.

(D) In an isosmotic salt solution, there is no diffusion of water into or out of the paramecia, so the contraction rate is zero.

25. A human kidney filters about 200 liters of blood each day. Approximately two liters of liquid and nutrient waste are excreted as urine. The remaining fluid and dissolved substances are reabsorbed and continue to circulate throughout the body. Antidiuretic hormone (ADH) is secreted in response to reduced plasma volume. ADH targets the collecting ducts in the kidney, stimulating the insertion of aquaporins into their plasma membranes and an increased reabsorption of water.

If ADH secretion is inhibited, which of the following would initially result?

(A) The number of aquaporins would increase in response to the inhibition of ADH.

(B) The person would decrease oral water intake to compensate for the inhibition of ADH.

(C) Blood filtration would increase to compensate for the lack of aquaporins.

(D) The person would produce greater amounts of dilute urine. ✓

2 Cell MCQ

26. A human kidney filters about 200 liters of blood each day. Approximately two liters of liquid and nutrient waste are excreted as urine. The remaining fluid and dissolved substances are reabsorbed and continue to circulate throughout the body. Antidiuretic hormone (ADH) is secreted in response to reduced plasma volume. ADH targets the collecting ducts in the kidney, stimulating the insertion of aquaporins into their plasma membranes and an increased reabsorption of water.

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(C) Blood filtration would increase to compensate for the lack of aquaporins.

(D) The person would produce greater amounts of dilute urine. ✓

27. All of the following cell components are found in prokaryotic cells EXCEPT

(A) DNA

(B) ribosomes

(C) cell membrane

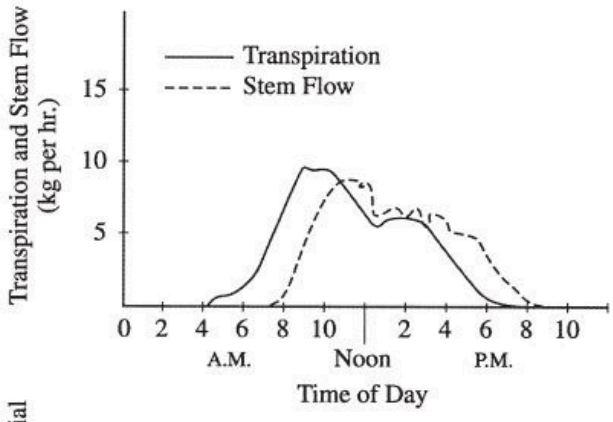
(D) nuclear envelope ✓

(E) enzymes

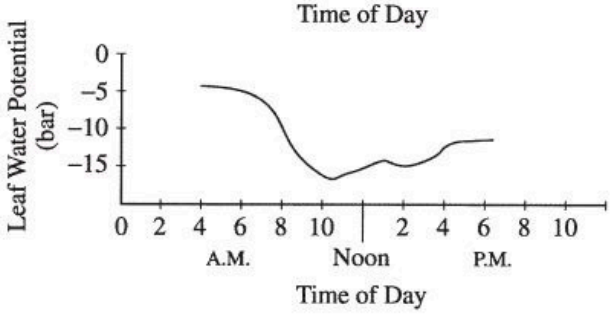
2 Cell MCQ

28. The rate of transpiration, the flow of water through the stem, and leaf water potential are measured in a tree during a 24-hour period under normal environmental conditions. The results from these measurements are shown in the graphs below.

Graph I



Graph II



All of the following changes would be likely to decrease the rate of transpiration at 8 A.M. EXCEPT

(A) causing the stomata to close

(B) increasing the humidity of the atmosphere

(C) increasing the water potential of the atmosphere

(D) increasing the water potential of the soil ✓

(E) placing the plant in total darkness

29. Researchers claimed that a particular organelle originated from a free-living prokaryotic cell that was engulfed by a larger cell, as shown in Figure 1.

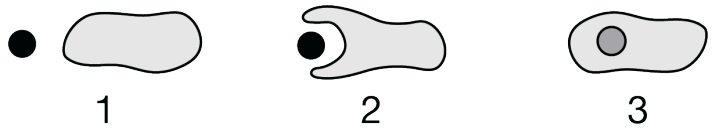


Figure 1. A model showing a cell engulfing a smaller cell

Which of the following provides evidence to best support the researchers' claim?

2 Cell MCQ

(A) The organelle has a phospholipid membrane.
(B) The organelle has protein in the membrane.
(C) The organelle has a double membrane. ✓
(D) The organelle has an internal aqueous environment that is similar to the cytosol of the larger cell.

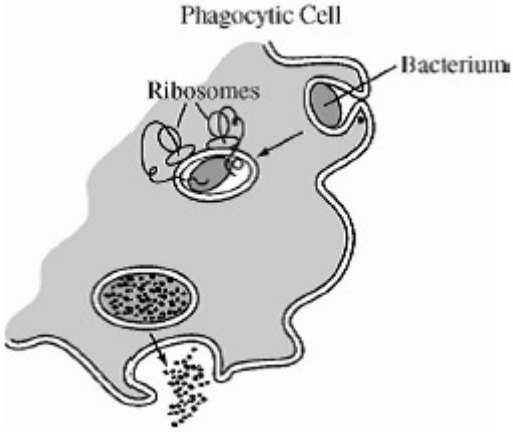
Answer C

Correct. The organelle's double membrane could have resulted from the process of endocytosis. Based on the claim, the organelle's inner membrane originated from the plasma membrane of the smaller cell and the organelle's outer membrane originated from the plasma membrane of the larger cell.

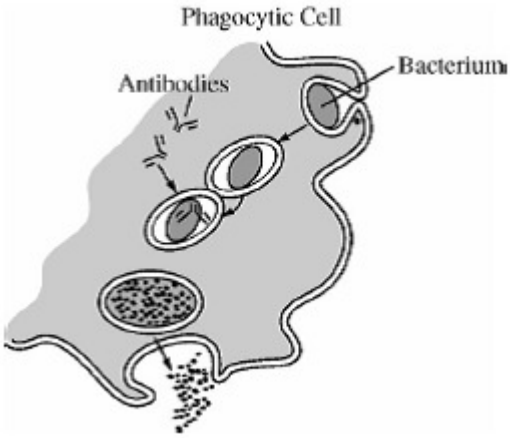
30. A pathogenic bacterium has been engulfed by a phagocytic cell as part of the nonspecific (innate) immune response. Which of the following illustrations best represents the response?

2 Cell MCQ

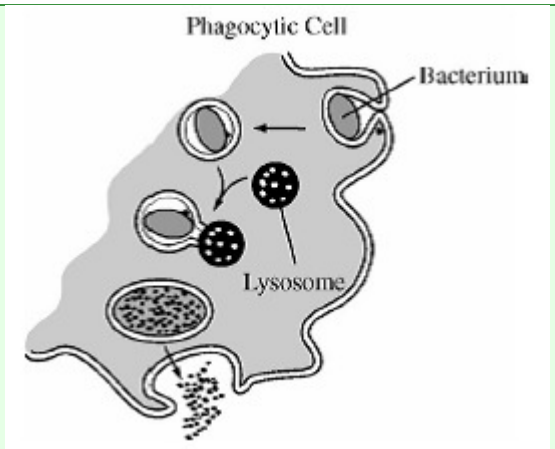
(A)



(B)

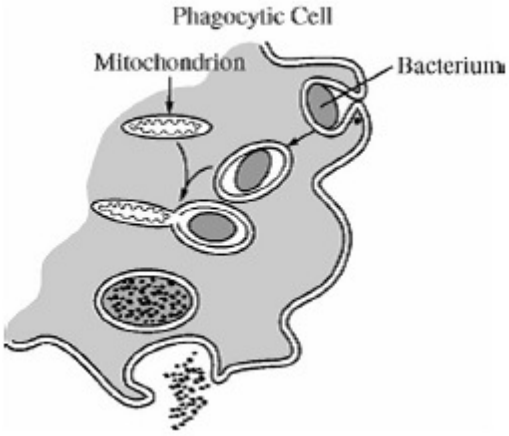


(C)



2 Cell MCQ

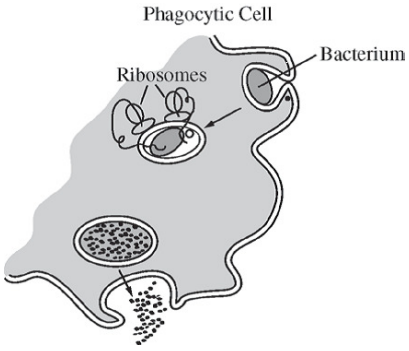
(D)



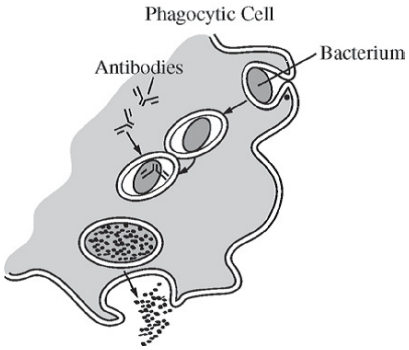
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2 Cell MCQ

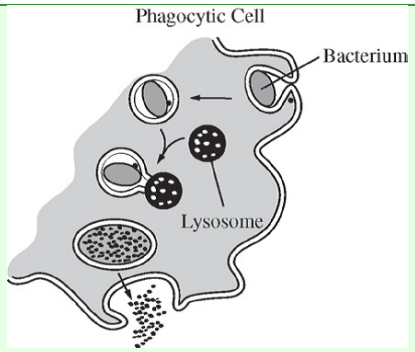
(A)



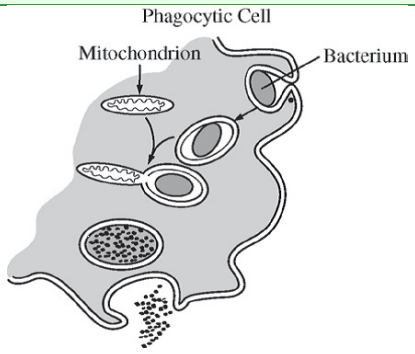
(B)



(C) ✓



(D)



Answer C

This option is correct. It demonstrates an understanding to interpret representations that describe a nonspecific immune response. In phagocytosis, a cell engulfs a bacterium by wrapping pseudopodia

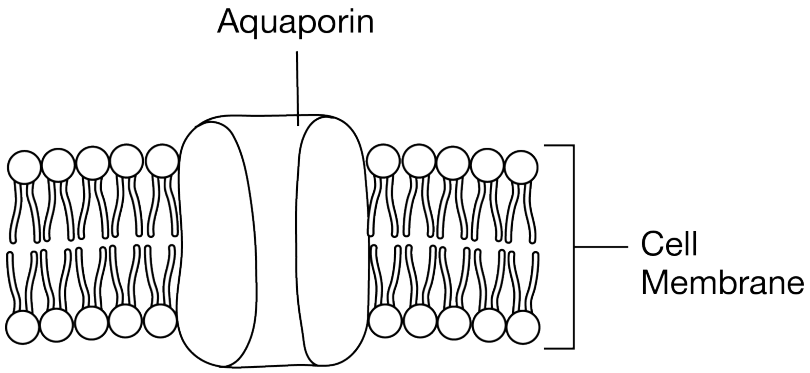
2 Cell MCQ

around it and packaging it within a membrane-enclosed vacuole. The bacterium is digested after the vacuole fuses with a lysosome containing hydrolytic enzymes. Digested particles can be exported out of the cell by exocytosis.

32. A prokaryotic cell has which of the following?

(A) Centrioles
(B) Lysosomes
(C) Plasma membrane ✓
(D) Mitochondria
(E) Endoplasmic reticulum

33.



Which statement best describes the effect on water transport across the cell membrane if the aquaporin in the figure ceases to function?

(A) Water molecules will no longer be able to move across the cell membrane.
(B) Water molecules will still be able to move across the cell membrane but at a slower rate. ✓
(C) Water molecules will only be able to enter the cell by active transport.
(D) Water molecules will move across the cell membrane at a faster rate without the aquaporin regulating their flow.

Answer B

Correct. The aquaporin facilitates water passage, but water can still move across the membrane by osmosis.

2 Cell MCQ

34. Aquaporins are channel proteins that facilitate the transport of water across the cell membrane. One group of researchers hypothesizes that without functional aquaporins, no water will be able to enter the cell. A different group proposes an alternative hypothesis, stating that even with nonfunctional aquaporins, a small amount of water will still cross the cell membrane. An experiment is set up in which plant cells with mutated (nonfunctional) aquaporins and plant cells with normally functioning aquaporins are both placed in distilled water.

Which of the following data would support the alternative hypothesis?

(A) Cells with functional aquaporins exhibit low turgor pressure and are hypertonic.

(B) Cells with functional aquaporins exhibit high turgor pressure and are hypotonic.

(C) Cells with mutated aquaporins exhibit an absence of turgor pressure and are completely plasmolyzed.

(D) Cells with mutated aquaporins exhibit moderate turgor pressure and are hypertonic. ✓

Answer D

Correct. Even with mutated aquaporins, a small amount of water is able to enter and leave the cell. Thus, turgor pressure is not lost and the alternative hypothesis is supported.

35. A study was conducted to understand the factors controlling the rate at which molecules or ions travel across cell membranes. An artificial membrane was created that was composed of a phospholipid bilayer only. The speed at which various substances crossed this membrane was measured.

Some substances can pass through an actual cell membrane much faster than they passed through the artificial membrane in this study. Which of the following statements best explains this finding?

(A) Actual cell membranes have a much thicker phospholipid bilayer than the artificial membrane does.

(B) Actual cell membranes have a variety of proteins embedded in the membrane that are absent in the artificial membrane. ✓

(C) Hydrophobic substances spend more time between the two layers of phospholipid in the artificial membrane than they do between the layers in an actual membrane.

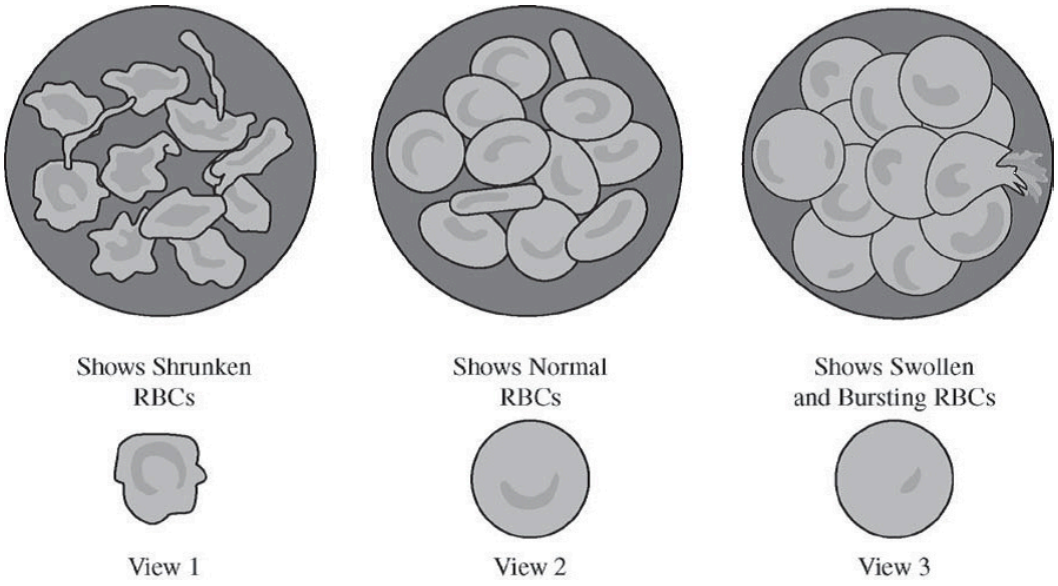
(D) Hydrophilic substances spend more time attached to the polar region of the phospholipids in the artificial membrane than they do attached to the polar region of the phospholipids in an actual membrane.

Answer B

Correct. Actual cell membranes have a variety of embedded proteins that increase the movement of substances across the membrane by both active and passive transport.

2 Cell MCQ

36. A sample of human blood was placed in a test tube containing a physiological saline solution (0.9% sodium chloride). This type of solution is often used intravenously to quickly rehydrate patients. A drop of the blood from the test tube was placed on a slide and red blood cells (RBCs) were observed under a microscope. Three possible outcomes are diagrammed below.



Which of the following best predicts which diagrammed microscope view the laboratory worker would see and best explains why?

(A) View 1 because RBC membranes are freely permeable to water

(B) View 2 because the RBCs use energy to allow sodium entry and to pump water out

(C) View 2 because the rate of water movement into the RBCs equals the rate of water movement out of the cells ✓

(D) View 3 because the sodium-potassium pumps in the RBC membranes use energy to keep the sodium out but allow water to freely flow into the cells

Answer C

This option is correct. The saline solution is isotonic relative to the RBC's cytoplasm. This will result in a dynamic equilibrium. View 2 shows a RBC in an isotonic environment because it would shrink in a hypertonic solution and swell in a hypotonic solution.

37. ATPase requires the use of ATP to move ions from one side of a cell membrane to another. The amino acid cysteine is important to the function of ATPase. A change in the cysteine prevents the ATPase from functioning.

Which of the following best describes what will happen to the movement of ions if there is a change in the cysteine of ATPase?

2 Cell MCQ

(A) Ions will move directly through the cell membrane.
(B) The electrochemical gradient will remain the same.
(C) The electrochemical gradient will change. ✓
(D) Ions will be unable to move through channel proteins.

38. If ATP breakdown (hydrolysis) is inhibited, which of the following types of movement across cell membranes is also inhibited?

(A) Movement of oxygen into a cell
(B) Movement of water through aquaporins
(C) Passage of a solute against its concentration gradient ✓
(D) Facilitated diffusion of a permeable substance

39. The antibiotic blasticidin S is a large molecule that can inhibit protein synthesis in mammalian cells. A researcher is investigating how blasticidin S enters mammalian cells. The researcher identified two proteins that might be associated with blasticidin S transport into the cell.

Which of the following claims most likely explains how the two proteins are directly involved in the antibiotic's ability to enter the cell?

(A) They form a receptor complex that selectively binds to blasticidin S in the cell cytoplasm.
(B) They are regulatory proteins that control the expression of other proteins involved in antibiotic transport.
(C) They are transport proteins that facilitate the movement of blasticidin S across the cell membrane. ✓
(D) They are enzymes that chemically modify blasticidin S to make it more soluble inside the cell.

40. Researchers have proposed a model of chloroplast evolution. According to the model, chloroplasts evolved from a small prokaryotic organism that was engulfed by an ancestral eukaryote. The engulfed prokaryote then formed an endosymbiotic relationship with the eukaryotic host.

Which of the following observations best supports the model?

(A) Chloroplasts are separated from other subcellular compartments by semipermeable membranes.
(B) Prokaryotic and eukaryotic organisms both acquire nutrients from the surrounding environment.
(C) Eukaryotes evolved after prokaryotes and have more complex structures.
(D) Chloroplasts and some prokaryotes share similar photosynthetic reactions. ✓

Answer D

Correct. Photosynthesis is a feature of only a limited number of organisms, including cyanobacteria, algae, and plants. The fact that this specialized process is similar in prokaryotes, such as cyanobacteria,

and in eukaryotes, such as algae and plants, provides evidence in support of the model.

41. Cholesterol is an important component of animal cell membranes. Cholesterol molecules are often delivered to body cells by the blood, which transports the molecules in the form of cholesterol-protein complexes. The complexes must be moved into the body cells before the cholesterol molecules can be incorporated into the phospholipid bilayers of cell membranes.

Based on the information presented, which of the following is the most likely explanation for a buildup of cholesterol molecules in the blood of an animal?

(A) The animal's body cells are defective in exocytosis.

(B) The animal's body cells are defective in endocytosis. ✓

(C) The animal's body cells are defective in cholesterol synthesis.

(D) The animal's body cells are defective in phospholipid synthesis.

Answer B

Correct. The cholesterol-protein complexes are most likely moved into the body cells by endocytosis. A defect in endocytosis is likely to result in a buildup of cholesterol in the blood.

42. When a substance moves across the plasma membrane along a concentration gradient at a rate faster than would be expected by simple diffusion alone but without the expenditure of metabolic energy, the process is best described as

(A) facilitated diffusion ✓

(B) osmosis

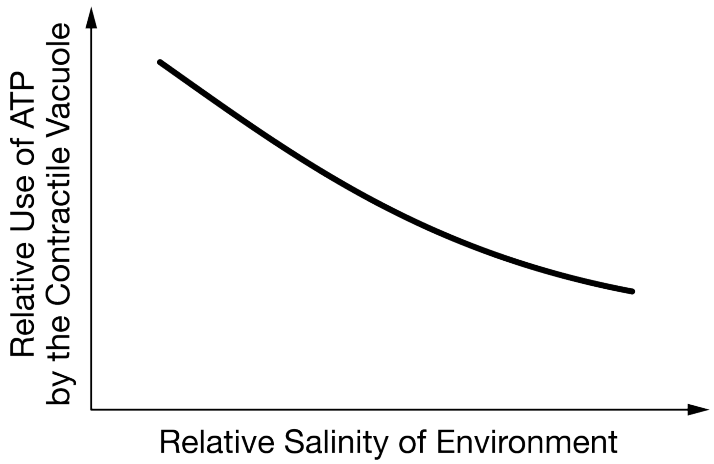
(C) active transport

(D) pinocytosis

(E) the work of an ion pump

2 Cell MCQ

43. Water is constantly diffusing into the cytosol of freshwater single-celled organisms. In order to maintain the proper solute concentrations in the cytosol, contractile vacuoles pump out the excess water. An experimenter placed single-celled organisms into various saline concentrations and recorded the ATP used by the contractile vacuole. The data are shown in the graph.



Of the following, which additional investigation can be used to determine when the cells are in an isotonic solution?

(A) Decreasing the salinity of the environment a little at a time until the ATP usage reaches a maximum

(B) Decreasing the salinity of the environment a little at a time until ATP usage reaches a minimum

(C) Increasing the salinity of the environment a little at a time until ATP usage reaches a maximum

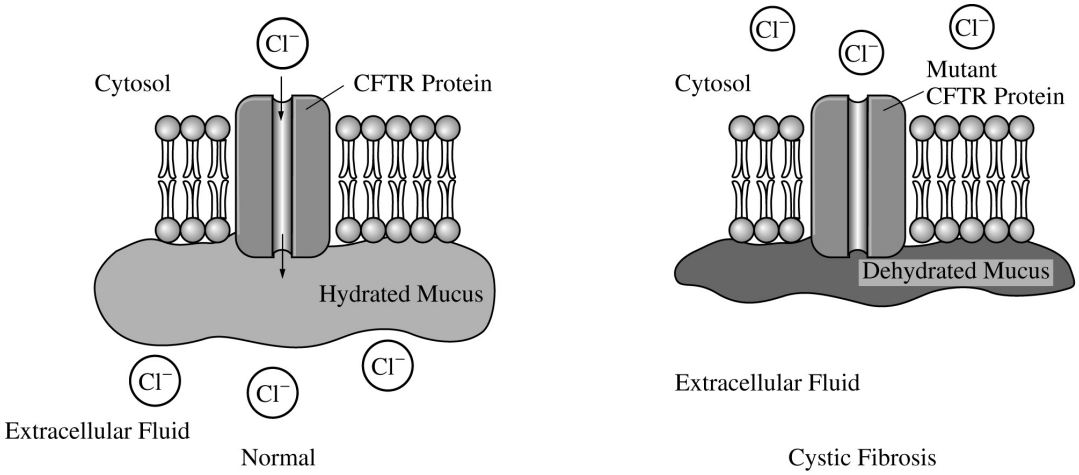
(D) Increasing the salinity of the environment a little at a time until the ATP usage reaches a minimum ✓

Answer D

Correct. If the environment is isotonic to the cell, the net flow of water due to diffusion is zero, and ATP is not used to pump water out of the cell.

2 Cell MCQ

44.



Cystic fibrosis is a recessively inherited disorder that results from a mutation in the gene encoding CFTR chloride ion channels located on the surface of many epithelial cells. As shown in the figure, the mutation prevents the normal movement of chloride ions from the cytosol of the cell to the extracellular fluid. As a consequence of the mutation, the mucus layer that is normally present on the surface of the cells becomes exceptionally dehydrated and viscous.

An answer to which of the following questions would provide the most information about the association between the CFTR mutation and the viscous mucus?

(A) Is the mucus also secreted from the cells through the CFTR proteins?

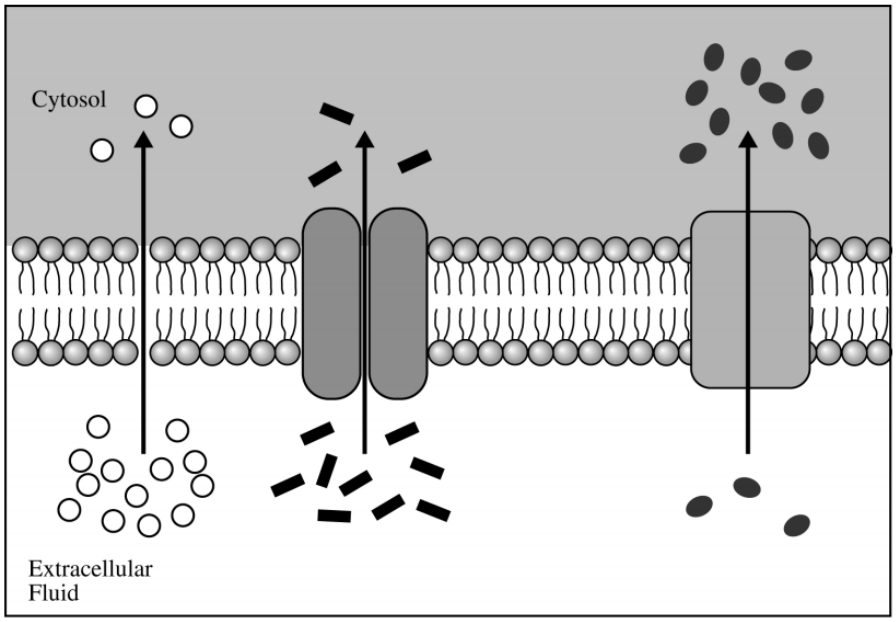
(B) How does the disrupted chloride movement affect the movement of sodium ions and water by the cell? ✓

(C) How does the mutation alter the structure of the CFTR proteins?

(D) What is the change in nucleotide sequence that results in the CFTR mutation?

2 Cell MCQ

45.



Which of the following scientific questions is most relevant to the model represented in the figure above?

(A) Is ATP required for the transportation of sugars across the outer mitochondrial membrane?

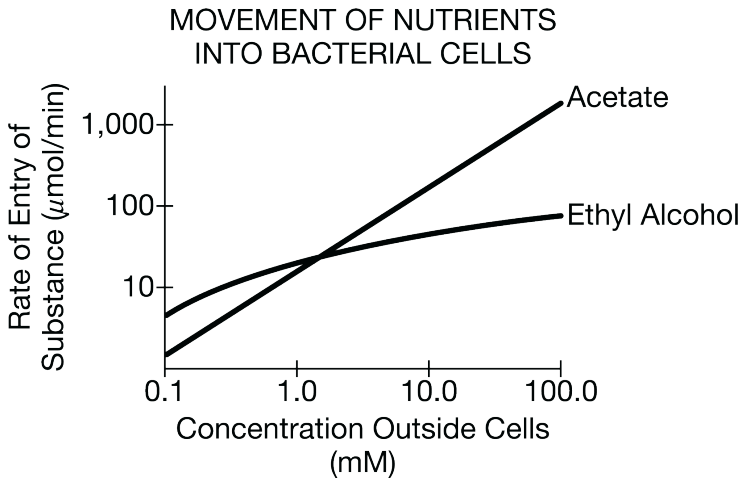
(B) Do the types of phospholipids in a membrane affect the rate at which molecules enter a cell by passive diffusion?

(C) Which molecular substance is actively transported across the plasma membrane? ✓

(D) How does temperature affect the movement of molecules into lysosomes?

2 Cell MCQ

46. Certain bacteria can use both ethyl alcohol and acetate as sources of nutrients. In an experiment where both nutrients are available to a bacterial population, the following results were obtained and graphed.



What additional procedure would best help determine whether these movements are due to active transport or to passive transport?

(A) Repeat the original experiment, but at three different temperatures. Compare the transport rates among the three temperatures.

(B) Repeat the original experiment, but add a substance known to block movement of molecules across aquaporins. Compare the rates on the two graphs.

(C) Use two additional treatments, one containing only ethyl alcohol and one containing only acetate. Compare the graphs of these two treatments with the original graph.

(D) Use two additional treatments, one containing only ethyl alcohol and one containing only acetate. Include a substance known to block ATP use by the plasma membrane. Compare the graphs of these two treatments to the original graph. ✓

Answer D

Correct. Active transport requires the availability of ATP. If the nutrients are actively transported, blocking ATP use should reduce the rate of transport along the whole concentration range. If these two graphs are the same as the original graph, the transport is passive. Separation of the nutrients eliminates the possibility that the two transport processes could interfere with each other.

2 Cell MCQ

47.

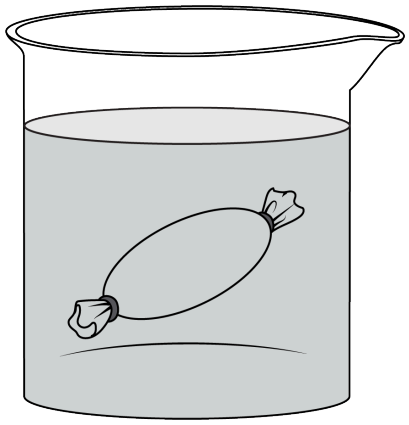


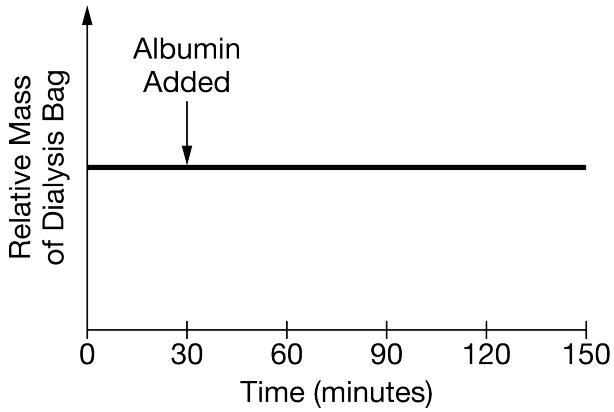
Figure 1

A student is using dialysis bags to model the effects of changing solute concentrations on cells. The student places one dialysis bag that contains $25\ \mathrm{mL}$ of distilled water into each of two beakers that are filled with $200\ \mathrm{mL}$ of distilled water. (Figure 1). The membrane of each dialysis bag membrane contains pores that allow small solutes such as monoatomic ions to pass through but are too small for anything larger to pass. After 30 minutes, $5\ \mathrm{mL}$ of a concentrated solution of albumin (a medium-sized, water-soluble protein) is added to one of the two beakers. Nothing is added to the other beaker. After two more hours at room temperature, the mass of each bag is determined. There is no change in the mass of the dialysis bag in the beaker to which no albumin was added.

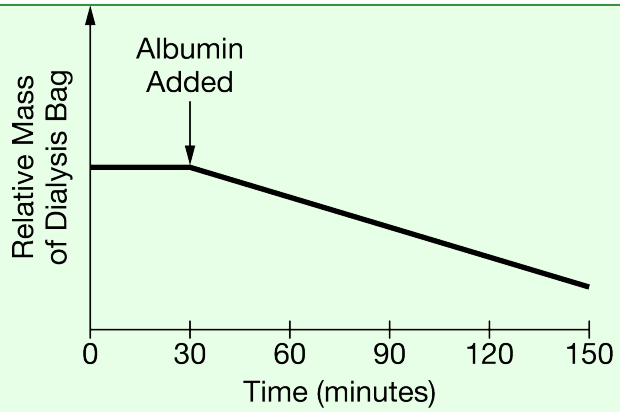
Which of the graphs below best represents the predicted change in mass over time of the dialysis bag in the beaker to which albumin was added?

2 Cell MCQ

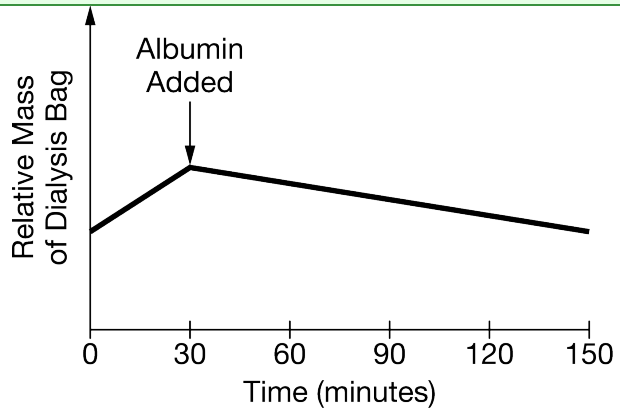
(A)



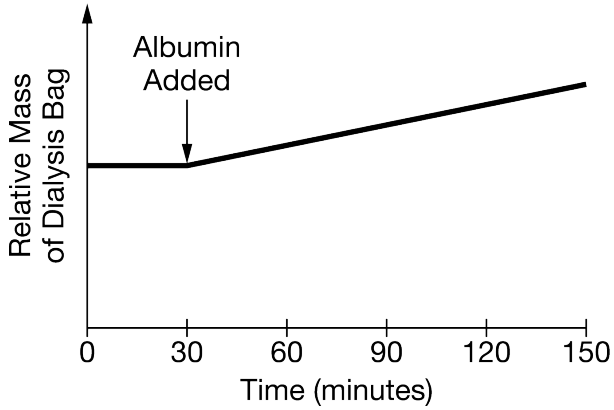
(B) ✓



(C)



(D)



2 Cell MCQ

48. Which of the following statements best supports the claim that certain organelles within eukaryotic cells evolved from free-living prokaryotic cells?

(A) The cytoplasm of both eukaryotes and prokaryotes is surrounded by a plasma membrane.

(B) Eukaryotes and prokaryotes both contain ribosomes, but the ribosomes of eukaryotes are more complex in structure than those of prokaryotes.

(C) Eukaryotes exchange segments of internal membranes between the endoplasmic reticulum and Golgi apparatus, but prokaryotes have no such internal membranes.

(D) Some organelles contain their own DNA that is more similar to prokaryotic DNA in structure and function than to the eukaryotic DNA found in the cell's nucleus. ✓

Answer D

Correct. The mitochondria and chloroplasts contain DNA more similar to prokaryote DNA in structure and function than to eukaryotic DNA. Eukaryotic DNA in the nucleus has additional structural proteins and a more complex process of protein synthesis. The most reasonable explanation involves the incorporation of one free-living prokaryotes that eventually integrated with the structures and processes of their eukaryotic hosts.

49. Which of the following is a characteristic of mitochondria and chloroplasts that supports the endosymbiotic theory?

(A) Both have bacteria-like polysaccharide cell walls.

(B) Both can reproduce on their own outside of the cell.

(C) Both contain DNA molecules. ✓

(D) Both contain endoplasmic reticulum and Golgi bodies.

(E) Both contain ribosomes that are identical to ribosomes of the eukaryotic cytoplasm.

2 Cell MCQ

50. Euglenids are single-cell eukaryotes that live in aquatic environments. The chloroplasts found inside euglenids are enveloped by three membranes, as represented in Figure 1. The inner membrane of euglenid chloroplasts resembles the thylakoid membrane.

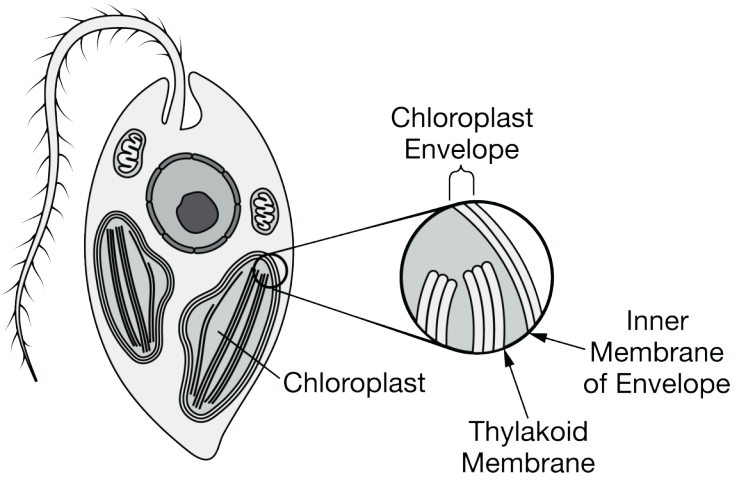


Figure 1. Simplified diagram of a euglenid, showing the structure of the chloroplast envelope

Which of the following claims about the origin of the euglenid chloroplast is best supported by the three-membrane structure of the envelope?

(A) It originated from the spontaneous assembly of organic molecules into a lipid bilayer inside a free-living prokaryote.

(B) It originated from the fusion of the plasma membranes of two different free-living photosynthetic prokaryotes.

(C) It originated from the incorporation of a photosynthetic prokaryote into a eukaryotic cell by a single endosymbiotic event.

(D) It originated from the incorporation of a photosynthetic prokaryote into a eukaryotic cell by two endosymbiotic events. ✓

Answer D

Correct. The three-membrane structure of the chloroplast envelope best supports the claim presented in this answer choice. The three-membrane structure most likely resulted from two endosymbiotic events, with the middle membrane originating from the first event and the outer membrane originating from the second event.

2 Cell MCQ

51. Gaucher disease is an inherited disorder in which cells of the body are unable to break down a particular type of lipid, resulting in a buildup of the lipid in some tissues and organs.

Based on the information provided, Gaucher disease results most directly from a defect in the function of which of the following organelles?

(A) The smooth endoplasmic reticulum

(B) The nucleus

(C) The lysosome ✓

(D) The mitochondrion

Answer C

Correct. The lysosome contains specific enzymes used to break down a variety of molecules and cellular waste products. A defect in the function of the lysosomal enzymes that are needed to break down lipids is the most direct cause of Gaucher disease.

52. Lysosomes digest food particles brought into a cell by endocytosis. After a vesicle containing food particles fuses with a lysosome, H^+ ions are transported into the lysosome from the cytosol. This significantly lowers the pH of the lysosome relative to the cytosol and activates the enzymes that digest the particles.

Which of the following best predicts what will happen to the lysosomal enzymes if the proteins that transport H^+ ions from the cytosol into the lysosome are damaged?

(A) The lysosomal enzymes will not become active, since there will be no active transport of H^+ ions. ✓

(B) The lysosomal enzymes will not become active, since H^+ ions will diffuse out of the lysosome.

(C) The lysosomal enzymes will become active, since facilitated diffusion will move H^+ ions into the lysosome.

(D) The lysosomal enzymes will become active, since passive diffusion will move H^+ ions into the lysosome.

Answer A

Correct. Since the pH of the lysosome has to become lower than that of the cytosol, active transport is needed to move H^+ ions into the lysosome, against the H^+ ion concentration gradient, to activate the enzymes.

2 Cell MCQ

53. Changing the shape or morphology of the mitochondrial inner membrane can change the efficiency of mitochondrial function.

Which of the following outcomes will most likely result from a change in the shape of the mitochondrial inner membrane from a highly folded surface to a smooth, flat surface?

(A) Mitochondria will become more efficient because the inner mitochondrial membrane will become more permeable to ions.

(B) Mitochondria will become more efficient because the total volume of the mitochondria will increase.

(C) Mitochondria will become less efficient because the inner mitochondrial membrane will become less permeable to large molecules.

(D) Mitochondria will become less efficient because the surface area of the inner mitochondrial membranes will decrease. ✓

Answer D

Correct. The change to the inner mitochondrial membrane will result in a decrease in the membrane's surface area, which will reduce the efficiency of mitochondrial function because there will be less surface area for reactions to occur.

2 Cell MCQ

Stickleback fish are found in both marine and freshwater habitats. The marine fish have no scales but have hardened, armorlike plates along their sides. The plates are thought to protect sticklebacks from certain predators.

In the late 1980s, sticklebacks from a marine population colonized Loberg Lake, a freshwater lake in Alaska. Starting in 1990, researchers sampled fish from the lake every four years and recorded the armor-plate phenotypes of the male sticklebacks in each sample. The armor-plate phenotypes were categorized as either complete (plates extending from head to tail), partial (plates extending from head to abdomen), or low (a few plates near the head only). The results are shown in the table below.

ARMOR-PLATE VARIATION IN THE STICKLEBACK POPULATION OF LOBERG LAKE

Percent of Males in the Sample with Each Armor-Plate Phenotype

Year	Low	Partial	Complete
1990	1%	2%	97%
1994	45%	14%	41%
1998	58%	16%	26%
2002	76%	15%	9%
2006	90%	6%	4%

54. Intact cells of two unknown cell types were placed into solutions with different concentrations of NaCl. Type I cells swelled and burst in the solution with the lowest concentration of NaCl. Type II cells swelled but did not burst in the solution with the lowest concentration of NaCl.

Which of the following descriptions of cell type I and cell type II are most consistent with the data?

2 Cell MCQ

(A)

Cell Type I	Cell Type II
Animal cell surrounded by a plasma membrane only	Plant cell surrounded by a plasma membrane and a cell wall

(B)

Cell Type I	Cell Type II
Plant cell surrounded by a plasma membrane only	Bacterial cell surrounded by a cell wall only

(C)

Cell Type I	Cell Type II
Plant cell surrounded by a plasma membrane only	Animal cell surrounded by a plasma membrane and a cell wall

(D)

Cell Type I	Cell Type II
Animal cell surrounded by a cell wall only	Bacterial cell surrounded by a plasma membrane only

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55. Which of the following statements accurately describes the facilitated diffusion of materials across a cell membrane?

(A) Energy is required to move charged substances across the membrane.

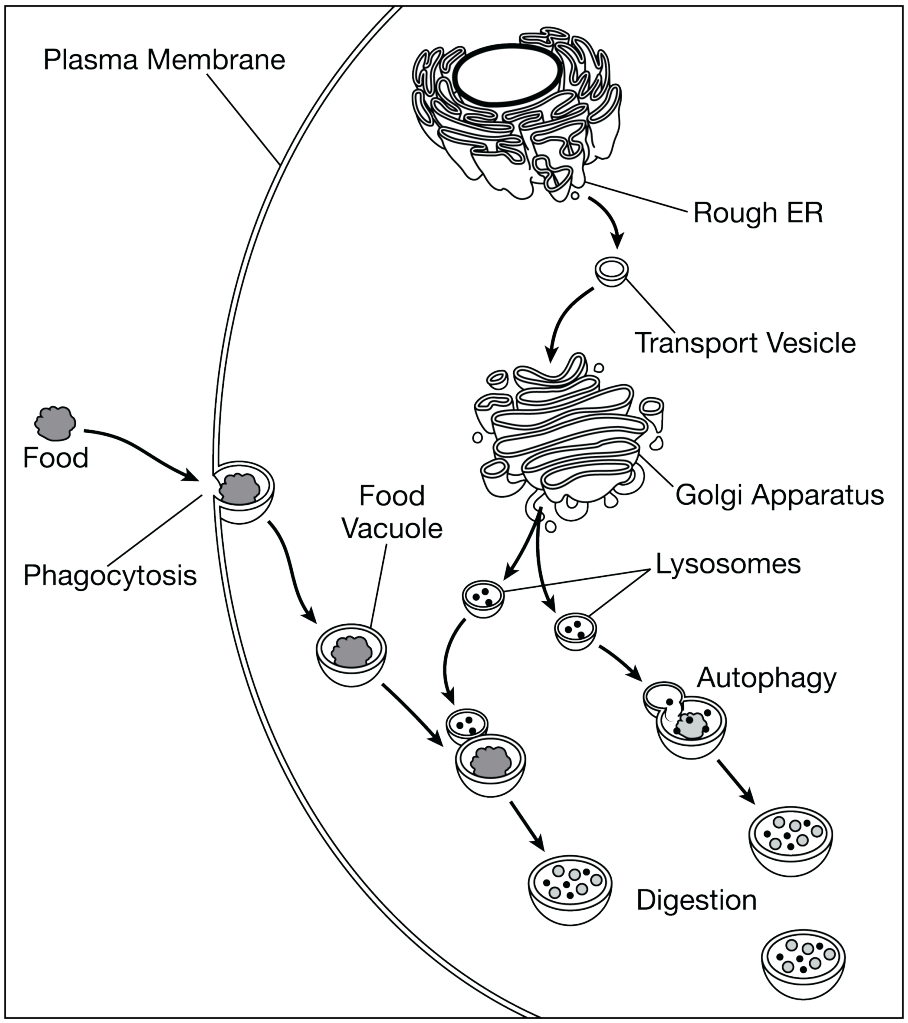
(B) Small, polar molecules require transport proteins to cross the membrane.

(C) Oxygen molecules move directly across the membrane.

(D) Ions move down their concentration gradient across the membrane.

2 Cell MCQ

56. The figure shows a process by which a cell might absorb food from its surrounding environment and break it down for use as a source of energy and matter. The process involves lysosomes, which are membrane-bound organelles that contain hydrolytic enzymes. Activation of the hydrolytic enzymes requires an acidic pH, and lysosomes maintain an internal acidic pH by using ion pumps.



Which of the following outcomes will most likely result from a loss of ion pump function in the cell's lysosomes?

(A) The internal pH of the lysosomes will decrease, which will prevent the activation of hydrolytic enzymes and interfere with the intracellular digestion of food.

(B) The internal pH of the lysosomes will increase, which will prevent the activation of hydrolytic enzymes and interfere with the intracellular digestion of food. ✓

(C) The internal pH of the lysosomes will decrease, which will activate hydrolytic enzymes and enhance the intracellular digestion of food.

(D) The internal pH of the lysosomes will increase, which will activate hydrolytic enzymes and enhance the intracellular digestion of food.

2 Cell MCQ

Answer B

Correct. The lysosomes rely on the ion pumps to maintain an internal acidic pH. A loss of ion pump function will most likely result in an increase in the internal pH of lysosomes from acidic to neutral, which will prevent the activation of hydrolytic enzymes and interfere with intracellular digestion of food.

57. Some viral infections can lead to the rupture of the lysosome membrane. Which prediction of the effect of this disruption of cellular compartmentalization is most likely correct?

(A) Enzymes will be released that will specifically target the virus.

(B) Cellular osmotic concentrations will change, preventing viral entry into the cell.

(C) Hydrolytic enzymes will be released, which will cause cell death. ✓

(D) Intracellular digestion of organic materials will increase, which will increase the energy available to the cell for fighting the virus.

Answer C

Correct. Hydrolytic enzymes will be released, resulting in cell death and preventing further viral reproduction.

2 Cell MCQ

58. A cell's membrane potential is maintained by the movement of ions into and out of the cell. A model showing the influence of membrane proteins on the movement of sodium (Na^+) and potassium (K^+) ions across the plasma membrane is presented in Figure 1.

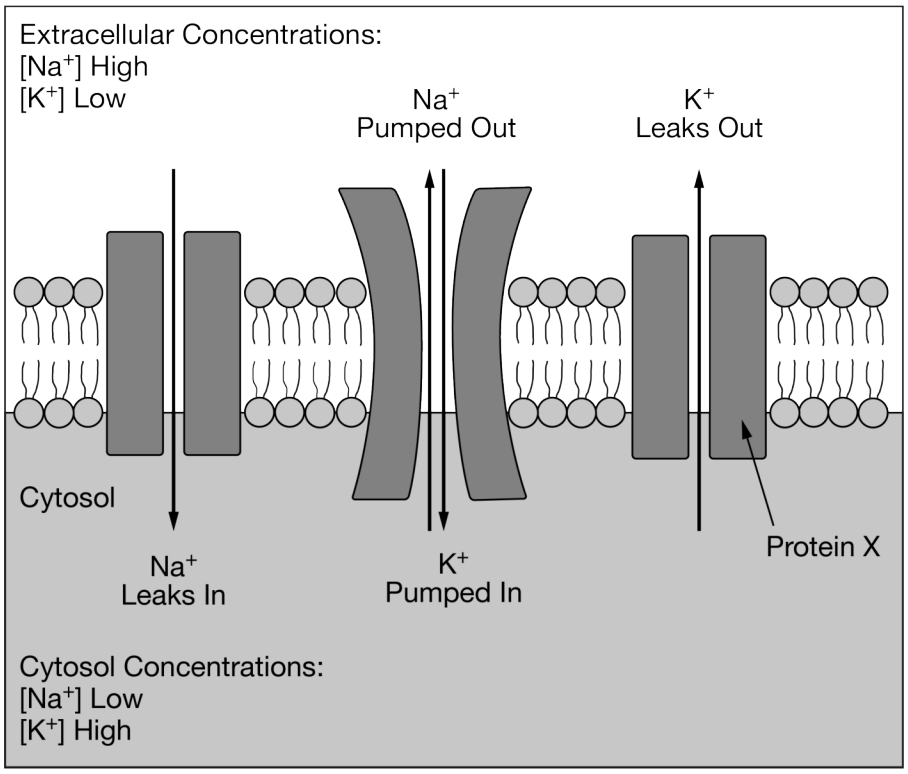


Figure 1. Section of a cell's plasma membrane, showing ion concentrations and membrane proteins

Based on the model presented in Figure 1, which of the following outcomes will most likely result from a loss of protein X function?

(A) The membrane potential will be disrupted by an increase in Na^+ concentration inside the cell.

(B) The membrane potential will be disrupted by an increase in K^+ concentration inside the cell. ✓

(C) The membrane potential will be maintained by the $Na^+ - K^+$ pump moving more K^+ ions into the cell.

(D) The membrane potential will be maintained by the diffusion of Na^+ ions into the cell.

Answer B

Correct. The intracellular K^+ concentration will increase because the $Na^+ - K^+$ pump will continue to

transport K^+ into the cell without any diffusing out of the cell. As a result, the cell's membrane potential will be disrupted.

59. Researchers investigate the transport of a certain protein into cells by endocytosis. In an experiment, the researchers incubate the cells in the presence of the protein and measure the amount of the protein that is absorbed into the cells over a five-minute period.

Based on their observations, what should the researchers do to further clarify how the availability of the protein outside the cells affects the rate of endocytosis of the protein?

(A) Incubate the cells in the absence of the protein.

(B) Incubate the cells in the presence of several different proteins.

(C) Incubate the cells in the presence of several different concentrations of the protein. ✓

(D) Incubate the cells in the presence of the protein for several different lengths of time.

Answer C

Correct. Changing the concentration of the protein will change the availability of the protein outside the cells.

60. Membrane-bound organelles have been an important component in the evolution of complex, multicellular organisms. Which of the following best summarizes an advantage of eukaryotic cells having internal membranes?

(A) Eukaryotic cells are able to reproduce faster because of the presence of organelles.

(B) Some organelles, such as mitochondria and chloroplasts, are similar to prokaryotic cells in structure.

(C) Organelles isolate specific reactions, increasing metabolic efficiency. ✓

(D) Compartmentalization leads to a higher mutation rate in DNA, which leads to more new species.

Answer C

This option is correct. It demonstrates understanding and provides an explanation of how internal membrane-bound organelles contribute to cell function by providing a favorable local environment for metabolic reactions, and by protecting the cell from potentially damaging metabolic reactions.

61. Mitochondria are found in most eukaryotic cells and contain their own DNA and ribosomes that are similar to those typical of many prokaryotic cells. Which of the following statements is justified by these observations?

2 Cell MCQ

(A) The mitochondrion is the only location in which eukaryotic cells can synthesize ATP.

(B) An ancestral cell most likely engulfed an aerobic prokaryote in a relationship that proved beneficial for both cells. ✓

(C) Mitochondrial membranes provide abundant surface area for reactions because of the infoldings called cristae.

(D) The mitochondrion plays a role in respiration, but it also stores the cell's extra mRNA.

62. The endosymbiont theory proposes a model for the evolution of mitochondria. According to the model, an ancestral eukaryote engulfed a small, free-living prokaryotic organism. The engulfed prokaryote then formed an endosymbiotic relationship with the eukaryotic host.

Which of the following observations best supports the model?

(A) Prokaryotes and eukaryotes acquire nutrients from the surrounding environment.

(B) Organelles such as mitochondria and the endoplasmic reticulum have membranes composed of phospholipids.

(C) Mitochondria and some prokaryotes share similar metabolic reactions that produce ATP. ✓

(D) Eukaryotes evolved after prokaryotes and have more complex structures.

Answer C

Correct. The observation that mitochondria and some prokaryotes share similar metabolic reactions that produce ATP supports the model that mitochondria are endosymbionts that originated from a free-living prokaryote.

63. Which of the following observations best supports the claim that mitochondria evolved from once-free-living prokaryotic cells by the process of endocytosis?

(A) Mitochondria produce ATP.

(B) Mitochondria contain proteins.

(C) Mitochondria exchange substances with the cytosol.

(D) Mitochondria are surrounded by a double membrane. ✓

Answer D

Correct. The double membranes of mitochondria provide evidence that an ancestor of mitochondria, which was most likely a type of free-living aerobic bacterium, was ingested via endocystosis by a primitive eukaryotic cell.

2 Cell MCQ

64. Which of the following statements best predicts the effect of increasing the permeability of the mitochondrial membranes to large molecules?

(A) ATP production will increase because of an increase in the rate at which proteins diffuse out of mitochondria.

(B) ATP production will increase because of an increase in the mixing of mitochondrial and cytosolic substances.

(C) ATP production will decrease because of an increase in the occurrence of uncontrolled chemical reactions. ✓

(D) ATP production will decrease because of an increase in the surface area of the mitochondrial membranes.

Answer C

Correct. ATP production will most likely decrease. Increasing the permeability of the mitochondrial membranes to large molecules will likely result in a variety of uncontrolled substrate-enzyme interactions that will disrupt normal mitochondrial function.

65. Which of the following accurately describes a model of active transport?

(A) A membrane channel protein facilitates the movement of molecules from an area of high concentration on the outside of the cell to an area of lower concentration on the inside.

(B) A cell in a hypotonic solution swells as water moves into it.

(C) A membrane protein binds a specific molecule outside of the cell and moves the molecule into the cell where there is already a high concentration of the molecule compared with the outside. ✓

(D) A membrane protein binds a specific molecule on the inside of the cell and moves the molecule to an area of lower concentration on the outside of the cell.

Answer C

Correct. Active transport uses a membrane protein to move a specific molecule up the concentration gradient from an area of low concentration to an area of high concentration.

66. The Na^+/K^+ pump requires ATP to move the ions from one side of a membrane to the other. Which of the following explains the role ATP plays in the pump's activity?

2 Cell MCQ

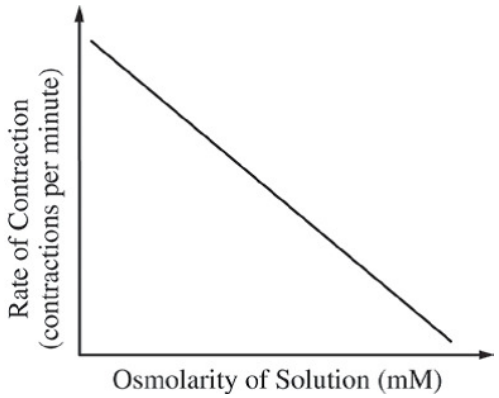
(A) ATP provides energy for the ions to move down the concentration gradient.

(B) ATP adds a phosphate to the K^+ ion so that it is not charged and can be moved by the pump.

(C) ATP provides energy for the ions to move against the concentration gradient. ✓

(D) ATP adds a phosphate to the Na^+ ion so that it is not charged and can be moved by the pump.

67.



Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experimental investigation, paramecia were placed in salt solutions of increasing osmolarity. The rate at which the contractile vacuole contracted to pump out excess water was determined and plotted against osmolarity of the solutions, as shown in the graph. Which of the following is the correct explanation for the data?

(A) At higher osmolarity, lower rates of contraction are required because more salt diffuses into the paramecia.

(B) The contraction rate increases as the osmolarity decreases because the amount of water entering the paramecia by osmosis increases. ✓

(C) The contractile vacuole is less efficient in solutions of high osmolarity because of the reduced amount of ATP produced from cellular respiration.

(D) In an isosmotic salt solution, there is no diffusion of water into or out of the paramecia, so the contraction rate is zero.

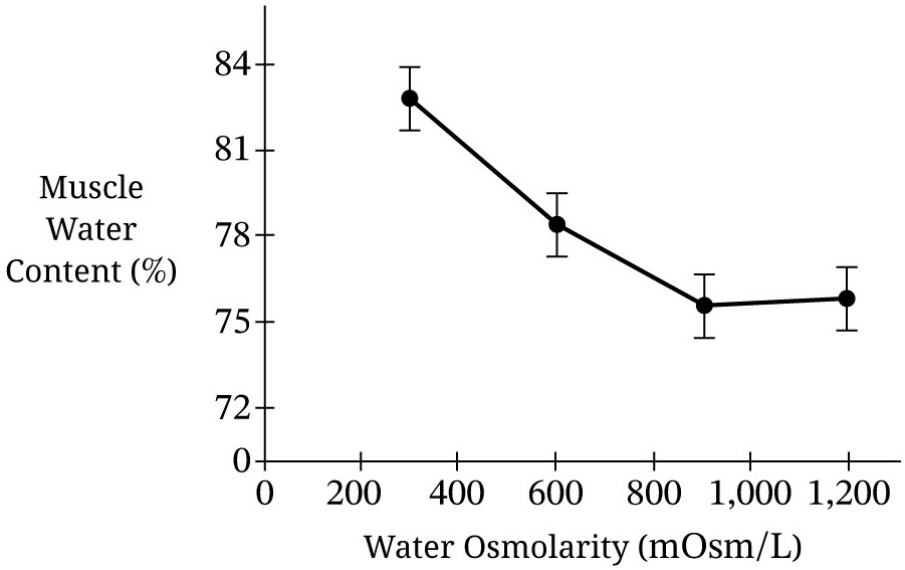
Answer B

This option is correct. The lower the concentration of solute, the larger the gradient between the paramecium and the solution. When the paramecium is hypertonic to the low concentration solution, water will enter the paramecium by osmosis. If the paramecium does not pump excess water back out, it will burst. As the solution gradient decreases, less water will enter, and, therefore, there will be a decreased need to pump excess water out.

2 Cell MCQ

68. Researchers studying osmoregulation in a specific crab species measured the change in water content within the crabs' muscles when the crabs were exposed to water with different salt concentrations (Figure 1).

Figure 1. Muscle water content in crabs at different saltwater osmolarities. Error bars represent $\pm 2SE_{\bar{x}}$



Which of the following claims about the crab's muscle water content is most likely supported by the data shown in Figure 1?

(A) The water content remains relatively constant across different levels of water osmolarity.

(B) The water content increases and then decreases as water osmolarity increases.

(C) The water content decreases as water osmolarity increases. ✓

(D) The water content increases as water osmolarity increases.

2 Cell MCQ

69. **TABLE 1. INTERNAL AND EXTERNAL WATER POTENTIALS OF FOUR DIFFERENT PARAMECIA**

Paramecium	Internal Ψ	External Ψ
1	-1.44 MPa	-0.08 MPa
2	-1.56 MPa	-1.01 MPa
3	-1.33 MPa	-1.76 MPa
4	-0.37 MPa	-1.73 MPa

A paramecium is a single-celled organism that contains one or more organelles called contractile vacuoles. In its freshwater environment, water may accumulate inside the paramecium and the contractile vacuoles. Contraction of the vacuoles removes excess water from the organism, thus preventing the paramecium from bursting. Table 1 provides internal and external water potentials (Ψ) for four different paramecia living in four different environments.

In which paramecium in Table 1 will the contractile vacuoles most likely have the highest contraction rate?

(A) Paramecium 1 ✓

(B) Paramecium 2

(C) Paramecium 3

(D) Paramecium 4

Answer A

Correct. Paramecium 1 has the largest difference of the four paramecia between the internal and external water potential, and the internal water potential is more negative than the external is. This will cause a net movement of water into the paramecium and will require the contractile vacuole to contract more in order to keep the paramecium from bursting.

70. Which of the following statements best explains the processes of passive and active transport?

2 Cell MCQ

(A) Passive transport is the net movement of substances down a concentration gradient that requires metabolic energy. Active transport is the movement of substances up a concentration gradient that does not require energy.

(B) Passive transport is the net movement of substances down a concentration gradient that does not require metabolic energy. Active transport is the movement of substances up a concentration gradient that requires energy. ✓

(C) Passive transport is the net movement of substances up a concentration gradient that requires metabolic energy. Active transport is the movement of substances down a concentration gradient that does not require metabolic energy.

(D) Passive transport is the net movement of substances up a concentration gradient that does not require metabolic energy. Active transport is the movement of substances down a concentration gradient that requires energy.

Answer B

Correct. Passive transport is the net movement of substances down (with) a concentration gradient that does not require metabolic energy and active transport is the movement of substances up (against) a concentration gradient that requires energy.

71. In an experiment, cells were isolated from an aquatic plant and suspended in pond water, a sucrose sugar solution, or distilled water. All of the cells were then viewed under a microscope. Compared with the cell in the pond water, the cell in the sugar solution appeared shriveled, and the cell in the distilled water appeared inflated. The results of the experiment are represented in Figure 1.

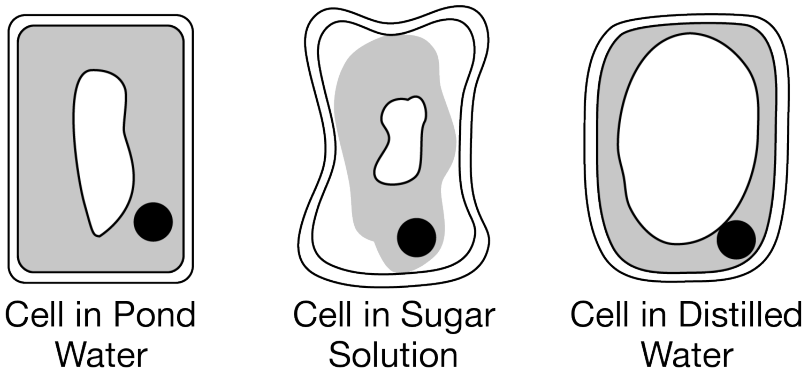


Figure 1. The results of an experiment using aquatic plant cells

Which of the following statements best explains the observations represented in Figure 1 ?

2 Cell MCQ

(A) There was a net movement of sucrose out of the cell suspended in the sugar solution and a net movement of sucrose into the cell suspended in the distilled water.

(B) There was a net movement of sucrose into the cell suspended in the sugar solution and a net movement of sucrose out of the cell suspended in the distilled water.

(C) There was a net movement of water out of the cell suspended in the sugar solution and a net movement of water into the cell suspended in the distilled water. ✓

(D) There was a net movement of water into the cell suspended in the sugar solution and a net movement of water out of the cell suspended in the distilled water.

Answer C

Correct. The plasma membrane that surrounds a living cell is typically permeable to water but not sucrose. In the experiment represented in Figure 1, there was a net movement of water out of the cell suspended in the sugar solution, causing the cell to appear shriveled, and a net movement of water into the cell suspended in the distilled water, causing the cell to appear inflated.

72. Which of the following are characteristic of both prokaryotic and eukaryotic cells?

(A) Cytoplasm and a well-defined nucleus surrounded by a membrane

(B) Membranous sites of ATP synthesis, Golgi complex, and ribosomes

(C) Mitochondria, nucleus, and ribosomes

(D) Cell wall, several chromosomes, and cytoplasm

(E) Cell membrane, ribosomes, DNA, and RNA ✓

73. Which of the following groups of cellular components are found in eukaryotic cells but not prokaryotic cells?

(A) Ribosomes, a nucleus, and chloroplasts

(B) Circular chromosomes, mitochondria, and an endoplasmic reticulum

(C) A nucleus, ribosomes, and cell walls

(D) An endoplasmic reticulum, mitochondria, and a nucleus ✓

2 Cell MCQ

74.

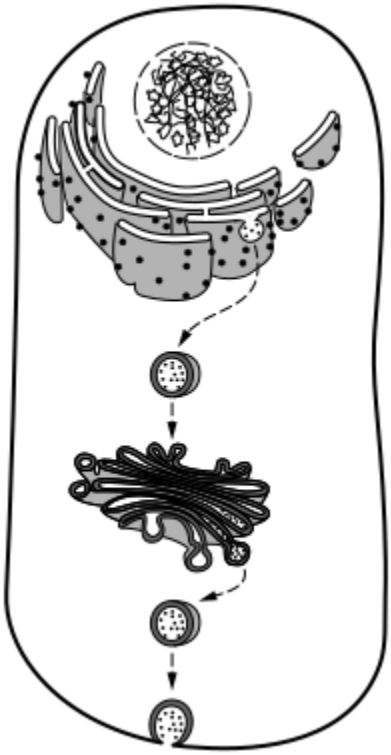


Figure 1. The path of insulin export from a cell

Insulin is a protein hormone that helps to regulate blood sugar levels. Figure 1 shows the path of insulin from its production in a cell to its export out of the cell.

Based on Figure 1, which of the following best describes how insulin leaves the cell?

(A) A transport protein carries insulin from the cytosol through the plasma membrane by facilitated diffusion.

(B) A transport protein in the plasma membrane moves insulin by active transport from the Golgi complex through the plasma membrane.

(C) A vesicle containing insulin moves through the plasma membrane by simple diffusion.

(D) A vesicle containing insulin fuses with the plasma membrane, enabling exocytosis of insulin. ✓

Answer D

Correct. Because insulin is secreted from the cell, it moves progressively through the components of the endomembrane system (rough endoplasmic reticulum and Golgi complex). This involves packaging insulin in a Golgi-derived transport vesicle, which then fuses with the plasma membrane, releasing

insulin by exocytosis.

75. The transport of a substance across a plasma membrane of a specific organelle requires energy. The rate at which the transport takes place also depends on temperature. A scientist isolated the specific organelle and then used the following treatments to determine the conditions that will result in the maximal transport. All treatments contained the extracted organelle and were maintained at $25^\circ\mathrm{C}$.

The data from this experiment indicate that maximal rate of transport of protein X at $25^\circ\mathrm{C}$ occurs at an ATP concentration of $1.0\ \mu\mathrm{m/mL}$.

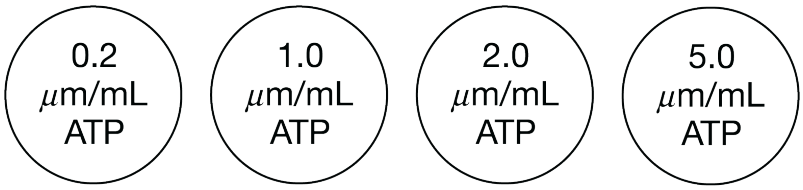


Figure 1. The four ATP concentrations used in the experiment

Which procedure should be done next to gather data needed to meet the scientist's objective?

(A) Incubate samples with the same four ATP concentrations at $30^\circ\mathrm{C}$.

(B) Incubate samples containing $5.0\ \mu\mathrm{m/mL}$ of ATP at four temperatures other than $25^\circ\mathrm{C}$.

(C) Incubate samples containing $1.0\ \mu\mathrm{m/mL}$ of ATP at four temperatures other than $25^\circ\mathrm{C}$. ✓

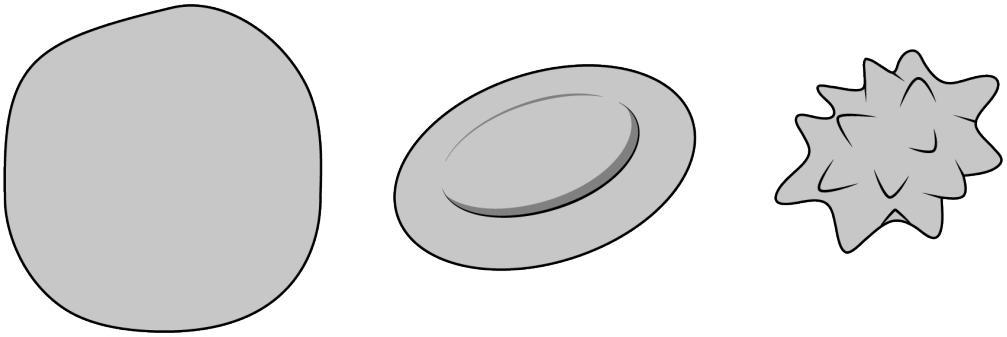
(D) Incubate samples containing $1.0\ \mu\mathrm{m/mL}$ of ATP at $25^\circ\mathrm{C}$ and determine the rate of transport for four other proteins.

Answer C

Correct. This procedure combines the optimal ATP concentration for transport with data for transport rates at four other temperatures.

2 Cell MCQ

76. Intravenous (IV) therapy is used for fluid replacement in instances of dehydration in humans and other animals. One type of IV fluid is essentially a saltwater solution. To determine the best concentration for therapy in people, a team of students is researching the effects of solutions of different salt concentrations on red blood cells. The following observations were made from three different red blood cell samples viewed under a microscope.



0.3% Saline (Cells swell) 0.9% Saline (Cells unchanged) 1.5% Saline (Cells shrink)

The team wants to extend the research project. What should the team of students do next to obtain data that are more conclusive?

(A) Repeat the process with other salt concentrations. ✓

(B) Develop a model to explain why the cells react differently to different salt concentrations.

(C) Repeat the process using red blood cells from other animals.

(D) Develop an experimental procedure that uses a stain that makes the organelles of red blood cells more visible.

Answer A

Correct. The figures illustrate the extremes of hypotonic (swollen cells) and hypertonic (shriveled cells) environments, so an extension would be to investigate the small range of acceptable salt concentrations in IV fluid solutions.

77. The salinity of a small inland lake has recently started to increase. Researchers are planning to study the lake over several decades to investigate how freshwater organisms survive significant changes in their natural habitat. Which of the following physiological mechanisms will the researchers most likely observe among the surviving organisms in the lake?

2 Cell MCQ

(A) Prokaryotic organisms will use various mechanisms to counteract swelling of cells as a result of increased water uptake.

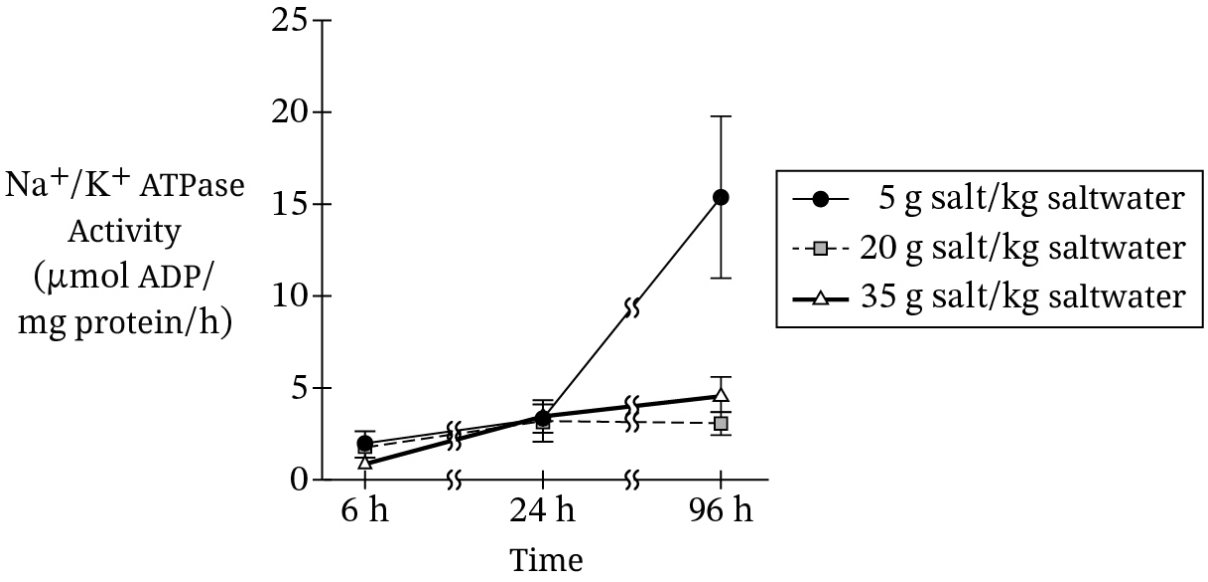
(B) Single-celled organisms will use various mechanisms to counteract the increased flow of water from cells to the environment. ✓

(C) Eukaryotic organisms will use various mechanisms to counteract the diffusion of positively charged ions across the cell membrane.

(D) Multicellular organisms will use various mechanisms to counteract the loss of cell adhesion as a result of calcium deficiencies.

78. A saltwater fish has specialized cells that contain Na^+/K^+ATPase to help with osmoregulation. Scientists studying this fish investigated the effect of the different concentrations of salt in the saltwater on the activity of Na^+/K^+ATPase between six and 96 hours (Figure 1).

Figure 1. Activity of Na^+/K^+ATPase at different saltwater concentrations over time. Error bars represent $\pm SE_{\overline{x}}$



Which of the following statements accurately supports the data shown in Figure 1?

(A) \(Na^+/K^+\)ATPase activity increases steadily as saltwater concentration and time increase.

(B) \(Na^+/K^+\)ATPase activity increases steadily as saltwater concentration decreases and time increases.

(C) \(Na^+/K^+\)ATPase activity stays approximately the same throughout the experiment and at all saltwater concentrations.

(D) \(Na^+/K^+\)ATPase activity increases only after 24 hours at the lowest saltwater concentration. ✓

Answer D

Correct. The data show that a significant increase in ATPase activity occurs only in the \(5\ \text{g}\ salt\

2 Cell MCQ

/\text{kg}\ water\) (lowest) concentration condition and only after the 24-hour mark. The other concentration treatments (20 and \(35\ \text{g}\ salt\ /\text{kg}\ water\)) show minimal changes over time.

79. Simple diffusion and facilitated diffusion are related in that both

(A) require protein carriers

(B) depend on a concentration gradient ✓

(C) occur via contractions of cytoskeletal elements attached to membrane proteins

(D) are endergonic processes and thus require the hydrolysis of ATP

(E) occur in eukaryotic cells but not in prokaryotic cells

80. Which of the following best supports the endosymbiotic theory?

(A) Mitochondria and chloroplasts have their own genomes. ✓

(B) Enzymes used in photosynthesis differ from those in cellular respiration in all eukaryotes.

(C) Bacteria and plants both have cell walls.

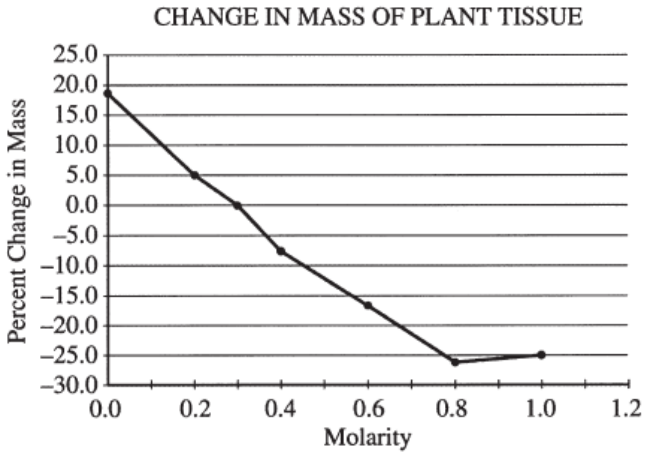
(D) Bacterial flagella are structurally different from eukaryotic flagella.

Answer A

Correct. The endosymbiotic theory claims that mitochondria and chloroplasts originated from early prokaryotic cells being engulfed and incorporated by another cell. These engulfed cells would have a double membrane due to endocytosis and would retain some of their original genomes. Mitochondria and chloroplasts possess both of these characteristics.

2 Cell MCQ

81. The following question refer to the graph below, which illustrates the percent change in the mass of pieces of plant tissue placed in solutions of different sucrose molarities.



The approximate molarity of the solution in which the mass of the plant pieces would not change is

(A) 0.0 *M*

(B) 0.1 *M*

(C) 0.3 *M* ✓

(D) 0.5 *M*

(E) 0.7 *M*

82. A magnesium sulfate solution taken orally can cause a net movement of water into the large intestine, which results from water molecules diffusing through aquaporins embedded in the cells of the intestinal lining.

By which of the following mechanisms do the water molecules most likely move into the large intestine?

(A) By passive transport from an area of low osmolarity to an area of high osmolarity ✓

(B) By passive transport from an area of high osmolarity to an area of low osmolarity

(C) By active transport from an area of low osmolarity to an area of high osmolarity

(D) By active transport from an area of high osmolarity to an area of low osmolarity

Answer A

Correct. Based on the information presented, the water molecules move through aquaporins by diffusing from an area of low osmolarity to an area of high osmolarity, which is an example of passive transport.

2 Cell MCQ

83.

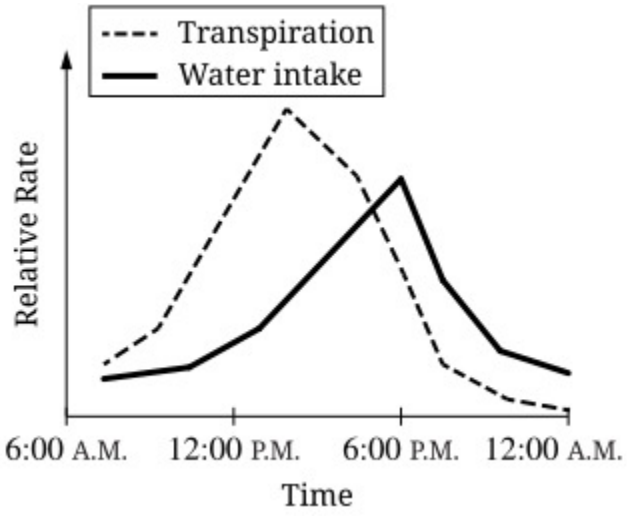


Figure 1. Relative rates of transpiration and water intake for a plant over 18 hours

Figure 1 shows the relative rates of transpiration (water loss) and water intake for a plant over an 18-hour period. Which of the following describes the most likely relationship between transpiration and water intake in the plant?

(A) Transpiration from the leaves of the plant stimulates subsequent water intake by the roots of the plant. ✓

(B) Transpiration and water intake peak at the same time of day.

(C) Water intake by the roots of the plant drives transpiration at the leaves of the plant.

(D) Transpiration and water intake have an inverse relationship.

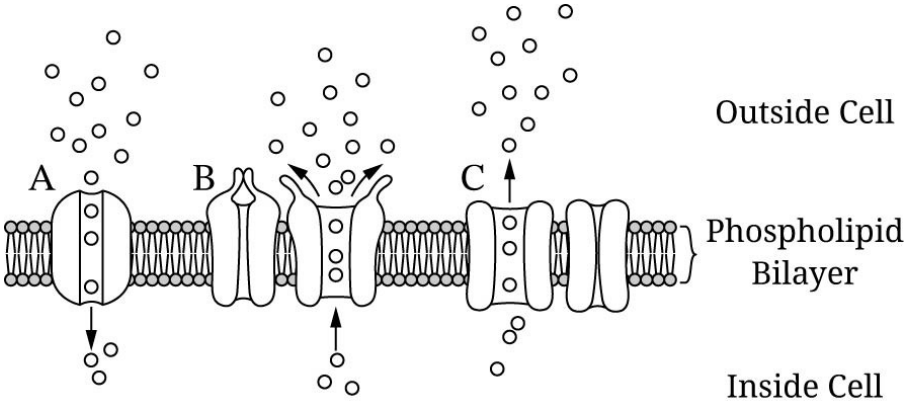
Answer A

Correct. In order for transpiration to be a stimulus for water intake, the rate of transpiration must peak before there is a peak in the rate of water intake. This relationship is supported by the data in Figure 1.

2 Cell MCQ

84. A researcher is studying three transport proteins (A, B, and C) in the membrane and has observed that protein A exhibits passive transport. Figure 1 shows the movement of solute particles in the different transport proteins. In the figure, two different conformations of the same protein are shown for protein B and protein C.

Figure 1. Transport proteins A, B, and C in the membrane



Based on Figure 1, which of the following experiments would best confirm that protein A utilizes passive transport, while proteins B and C do not?

(A) Measuring the rate of solute transport through each protein at various temperatures.

(B) Comparing the ATP consumption of cells expressing each protein type during solute transport. ✓

(C) Analyzing the amino acid sequences of all three proteins to identify structural similarities.

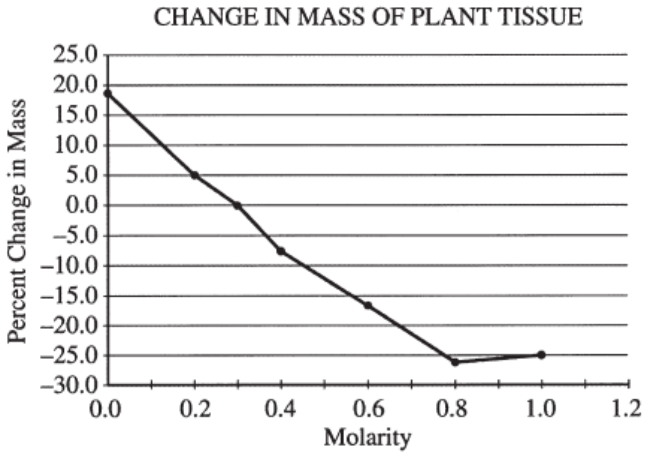
(D) Determining the direction and rate of solute movement through each protein.

Answer B

Correct. Since passive transport (protein A) does not require energy input while active transport (proteins B and C) requires ATP, measuring ATP consumption would clearly differentiate between these mechanisms. Cells expressing protein A would show no additional ATP consumption during transport.

2 Cell MCQ

85. The following question refer to the graph below, which illustrates the percent change in the mass of pieces of plant tissue placed in solutions of different sucrose molarities.



Water enters and leaves the plant cells primarily by

(A) endocytosis

(B) phagocytosis

(C) osmosis ✓

(D) active transport

(E) facilitated diffusion

86. Water is transported from the soil into the roots of a plant, and then up the stem to the leaves. A student determines that the water potential of root cells in the plant is -0.2 MPa. Which of the following claims is best supported by this measurement?

(A) The water potential of the soil will be higher than -0.2 MPa, whereas the water potential of the leaf will be lower than -0.2 MPa. ✓

(B) The water potential of the soil will be lower than -0.2 MPa, whereas the water potential of the leaf will be higher than -0.2 MPa.

(C) The water potential of both the soil and the leaf will be higher than -0.2 MPa.

(D) The water potential of both the soil and the leaf will be equal to -0.2 MPa.

Answer A

Correct. Water will move toward a region with lower water potential. If the water potential of the soil is greater than that of the roots and the water potential of the leaves is less than that of the roots, then water will move from the soil into the roots, up the stem, and to the leaves.

2 Cell MCQ

87. Which of the following provides the weakest evidence that mitochondria were once free-living prokaryotes?

(A) Mitochondrial ribosomes resemble those of prokaryotes.

(B) Mitochondria have DNA that is circular and does not have associated protein.

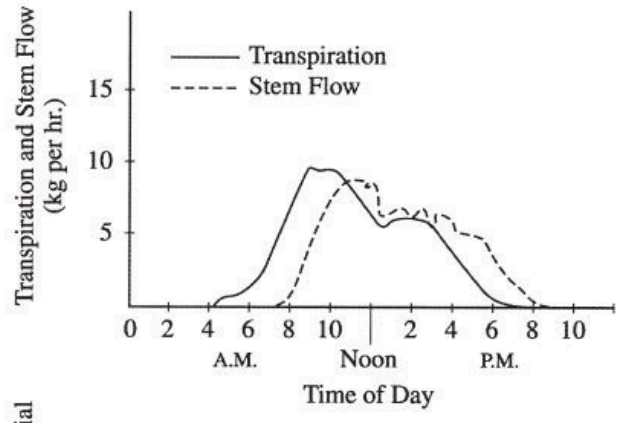
(C) Enzyme pathways on mitochondrial membranes resemble those found on modern prokaryote membranes.

(D) Mitochondria reproduce by a process similar to binary fission.

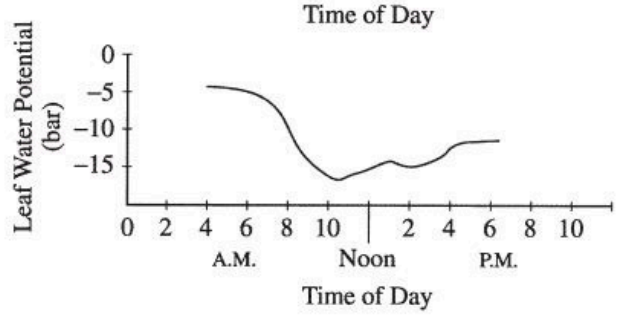
(E) Mitochondria and prokaryotes both are found in a variety of sizes. ✓

88. The rate of transpiration, the flow of water through the stem, and leaf water potential are measured in a tree during a 24-hour period under normal environmental conditions. The results from these measurements are shown in the graphs below.

Graph I



Graph II



What can be deduced from graph I?

(A) The rate of transpiration is constant through-out this 24-hour period.

(B) There is no correlation between the rate of transpiration and the flow of water through the stem.

(C) The maximal flow of water through the stem lags behind the maximal rate of transpiration. ✓

(D) The rate of transpiration never reaches zero.

(E) At noon, the rate of transpiration exceeds the rate of water flow through a stem.

89. Which of the following best describes an advantage that eukaryote organisms have over prokaryote organisms?

2 Cell MCQ

(A) Prokaryotes lack a cell membrane and therefore are unable to control what enters or exits the cell.

(B) Eukaryotes have a nuclear envelope separating their DNA from the rest of the cell, which increases the likelihood of advantageous mutations.

(C) Eukaryotes have mitochondria and chloroplasts that contain their own genome, which allows the cells to reproduce more rapidly.

(D) Eukaryotes have organelles that allow for compartmentalization of cellular processes, which increases the efficiency of those processes. ✓

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

In the first step of an experiment, rat liver cells were exposed for 5 minutes to amino acids labeled with a radioactive isotope. The cells were then washed to stop any further incorporation of radioactive amino acids. The cells were sampled periodically thereafter, and the radioactivity of a certain protein (protein X) was measured in various cell components, as shown below.

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

90. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

91. The same procedure was repeated with a second protein (protein Y). If the function of Y is protein processing and packaging, the most radioactivity would be found in the

(A) Golgi apparatus ✓

(B) Nucleus

(C) Mitochondria

(D) Lysosomes

(E) Cytoplasm

92. Which of the following correctly shows the order in which protein X moves through the cell?

2 Cell MCQ

(A) Endoplasmic reticulum→Golgi apparatus→lysosomes ✓

(B) Endoplasmic reticulum→Golgi apparatus→nucleus→cytoplasm→lysosomes

(C) Extracellular space→mitochondria→cytoplasm

(D) Golgi apparatus→cytoplasm→lysosomes

(E) Cytoplasm→mitochondria→extracellular space

93. In graphing the data from this experiment, the sampling time would be shown on the *x*-axis because

(A) cell components were shown on the *y*-axis

(B) radioactivity level shows the most variation

(C) time values are always located on the horizontal axis

(D) all quantitative values are represented on the *x*-axis

(E) sampling time is the independent variable ✓

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94. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

(D) Polar heads of phospholipids give plasma membranes less stability.

95. Which of the following processes is most likely to occur as a result of an animal cell receiving a signal to initiate apoptosis?

(A) Ribosomes will translate mRNA to produce proteins.

(B) Vesicles will release extracellular growth factors via exocytosis.

(C) Lysosomes will release digestive enzymes into the cytosol. ✓

(D) Vacuoles will fuse with the cellular membrane.

96. Which of the following organelles modifies and packages for secretion the materials produced by the ribosomes?

(A) The chloroplast

(B) The Golgi apparatus ✓

(C) The nucleus

(D) The nucleolus

(E) The mitochondrion

2 Cell MCQ

97. All eukaryotic cells contain at least one Golgi complex, typically located in the cytoplasm and near the endoplasmic reticulum.

Which of the following best describes a process that occurs within the Golgi complex?

(A) Enzymatic modification of newly synthesized integral membrane proteins ✓

(B) Synthesis of cytosolic proteins based on the nucleotide sequences of mRNAs

(C) Degradation of proteins by hydrolytic enzymes contained within the complex

(D) Synthesis of various types of lipids

Answer A

Correct. The Golgi complex is a membrane-bound structure to which some proteins are moved from the rough endoplasmic reticulum, where they are enzymatically modified before being sent to a cellular membrane.

98. Organelles such as mitochondria and the endoplasmic reticulum have membranes that compartmentalize reactions and other metabolic processes. To function properly, the organelles must move substances across their membranes.

Which of the following statements describes a feature shared by mitochondria and the endoplasmic reticulum that increases the efficiency of their basic functions?

(A) They have rigid, nonfluid membranes.

(B) They have highly folded membranes. ✓

(C) They have membranes composed of many carbohydrates.

(D) They have double membranes, with one membrane enclosed within the other.

Answer B

Correct. Compared with a smooth membrane, a highly folded membrane has a greater surface area that increases the rate of material transport across the membrane and provides more support for membrane-associated processes.

2 Cell MCQ

99. A cell is treated with a drug that prevents the formation of new lysosomes. The cell continues to transcribe the genes that code for the hydrolytic enzymes that are normally found in lysosomes and continues to translate the mRNAs for those proteins on membrane-bound ribosomes.

The hydrolytic enzymes are most likely to accumulate in which of the following cellular structures?

(A) Nucleus
(B) Mitochondrion
(C) Smooth endoplasmic reticulum
(D) Golgi complex ✓

100. Which of the following best describes a cell that is likely to have an extensive Golgi complex?

(A) A muscle cell that uses the energy stored in glucose molecules
(B) An intestinal cell that requires a large surface area
(C) A plant cell that uses chemiosmosis
(D) A stomach cell that produces and releases digestive enzymes ✓

Answer D

Correct. Stomach cells that produce and secrete digestive enzymes require an extensive Golgi complex because this organelle is responsible for processing, packaging, and preparing proteins, such as digestive enzymes, for secretion from the cell.

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following groups of biological compounds.

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

101. Synthesized at the ribosome

2 Cell MCQ

(A) Proteins ✓
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

---

102. Contains hydrolytic enzymes associated with the intracellular digestion of macromolecules

(A) Centriole
(B) Lysosome ✓
(C) Nucleolus
(D) Peroxisome
(E) Ribosome

103. Liver cells manufacture glycoproteins, while adipose cells store fat. Which of the following subcellular structures is likely to be more prominent in liver cells than in adipose cells?

(A) Nucleus
(B) Golgi apparatus ✓
(C) Cytoskeleton
(D) Plasma membrane

104. Which of the following best explains why hydrolytic enzymes are typically transported to the lysosome after they are synthesized by ribosomes?

(A) These enzymes are essential for breaking down organic molecules. ✓
(B) These enzymes function most efficiently at low concentrations.
(C) These enzymes require a continuous input of ATP to function.
(D) These enzymes require post-translational modifications to be bound to the cell membrane.

105. Prokaryotic and eukaryotic cells generally have which of the following features in common?

(A) A membrane-bound nucleus
(B) A cell wall made of cellulose
(C) Ribosomes ✓
(D) Flagella or cilia that contain microtubules
(E) Linear chromosomes made of DNA and protein

2 Cell MCQ

106. Muscle cells have high ATP demands. Which of the following is a scientific claim about how the structure of the mitochondria in muscle cells should be different than it is in other cells because of the high energy demands of mitochondria?

(A) The inner membrane of the mitochondria in muscle cells should have more folds to increase the surface area, allowing more ATP to be synthesized. ✓

(B) The inner membrane of the mitochondria in muscle cells should be more permeable to large enzymes, allowing the same reactions to occur in both compartments of the mitochondria.

(C) The outer membrane of the mitochondria in muscle cells should be thicker, allowing more rapid diffusion of molecules into the mitochondria.

(D) The outer membrane of the mitochondria of muscle cells should have more folds, increasing the surface area for faster diffusion of molecules from the cytoplasm.

Answer A

Correct. The amount of ATP produced is related to the number of ATP synthase molecules located on the inner mitochondria membrane. Increasing the folds increases the number of ATP generation sites and results in more ATP production.

107. Which of the following cellular deficiencies would most likely be related to mutations in mitochondrial proteins?

(A) The cell is unable to synthesize most proteins required for normal cell functions.

(B) The cell is unable to break down toxic materials and would accumulate large volumes of these materials.

(C) The cell is able to synthesize proteins, but the proteins would not fold properly and would not contain the correct molecular tags for export from the cell.

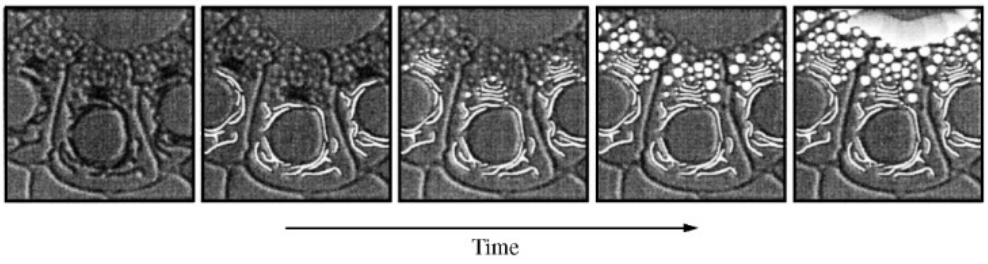
(D) The cell is unable to complete reactions related to electron transport and ATP production. ✓

Answer D

Correct. The electron transport chain and ATP production are associated with proteins in the inner membrane of the mitochondrion. Nonfunctional proteins in the mitochondrion are likely to result in reduced ATP production.

2 Cell MCQ

108.



The figure above shows a series of microscope images taken over time of ovalbumin production in chick oviduct cells in response to stimulation with estrogen. The ovalbumin protein was detected using a fluorescent dye as shown by the white areas in the time-lapse sequence.

The microscope images indicate that ovalbumin

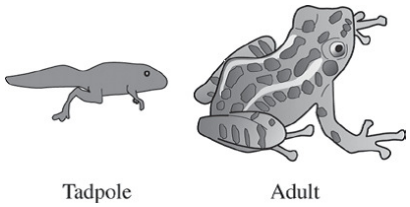
(A) is synthesized on soluble ribosomes and packaged into lysosomes, from which it is secreted into the oviduct

(B) is synthesized on ribosomes bound to the rough endoplasmic reticulum, and then secreted directly into the oviduct

(C) crosses the endoplasmic reticulum membrane as it is translated, then moves to the Golgi apparatus, then to vesicles from which it is secreted ✓

(D) crosses the Golgi apparatus membrane post-translationally, is taken up by the Golgi apparatus and endoplasmic reticulum, then is packaged into secretory vesicles

109.



The last part of the metamorphosis of a tadpole to an adult frog results in the disappearance of the tail. This stage of development most likely occurs by

(A) cells of the tail dying and the nutrients being absorbed and reused by the body ✓

(B) shedding of the tail so energy is not spent on maintenance of an unneeded part

(C) bilateral division of the tail and fusion with the developing hind limbs

(D) individual cells of the tail migrating to the developing gonads

Answer A

This option is correct. It demonstrates understanding of dynamic homeostasis in the reuse of nutrients by the body in development.

2 Cell MCQ

110. The organelle that is a major producer of ATP and is found in both heterotrophs and autotrophs is the

(A) chloroplast

(B) nucleus

(C) ribosome

(D) Golgi apparatus

(E) mitochondrion ✓

111. Which of the following claims is scientifically accurate and consistent with an observation that a decrease in lysosome production within a cell leads to a decline in mitochondrial activity?

(A) A lack of lysosomes will cause a decrease in the synthesis of enzymes necessary for cellular respiration.

(B) Fewer lysosomes will be available to break down macromolecules to provide the necessary nutrients for cellular respiration. ✓

(C) Fewer lysosomes will be available to store materials required for the functioning of the mitochondria.

(D) Lysosomes will not be available to modify proteins so that they are targeted to the mitochondria.

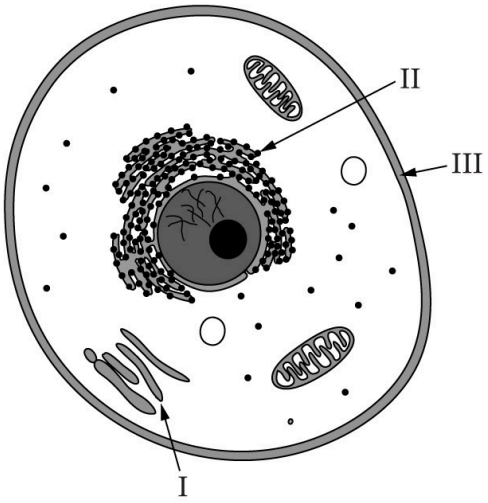
Answer B

Correct. A function of the lysosomes is the breakdown and recycling of macronutrients, such as carbohydrates. Products of carbohydrate breakdown such as glucose are utilized by the mitochondria in the process of cell respiration.

2 Cell MCQ

112. Figure 1 shows three labeled organelles (I, II, III) in a eukaryotic cell.

Figure 1. A eukaryotic cell



Which of the following is a claim that could be made about the relationship between organelles I, II, and III in Figure 1?

(A) All three organelles play a direct role in the production, modification, and transport of proteins that are released from the cell. ✓

(B) All three organelles play a direct role in the breakdown of materials that enter the cell.

(C) Organelles I and II synthesize macromolecules that are then transported by organelle III to the electron transport chain to become part of cellular respiration.

(D) Organelles II and III synthesize and transport proteins that are then digested into amino acids by organelle I.

113. Cells contain smaller components called organelles that are necessary for a cell's survival. Organelle functions have often been compared to components of larger systems.

Which of the following functional differences between the rough and smooth endoplasmic reticulum (ER) is explained by the structural differences between them?

(A) Rough ER breaks down toxic substances, and smooth ER only transports them out of the cell.

(B) Rough ER can synthesize and package lipids for export, and smooth ER cannot.

(C) Rough ER can produce ATP, and smooth ER cannot.

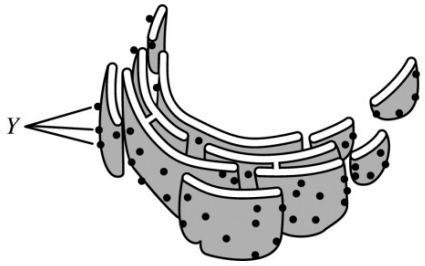
(D) Rough ER can synthesize and package proteins for export, and smooth ER cannot. ✓

2 Cell MCQ

Answer D

Correct. The ribosomes associated with rough ER synthesize proteins that are then exported from the cell.

114.



The figure above represents a rough endoplasmic reticulum. Which of the following best describes the role of the structure labeled *Y*?

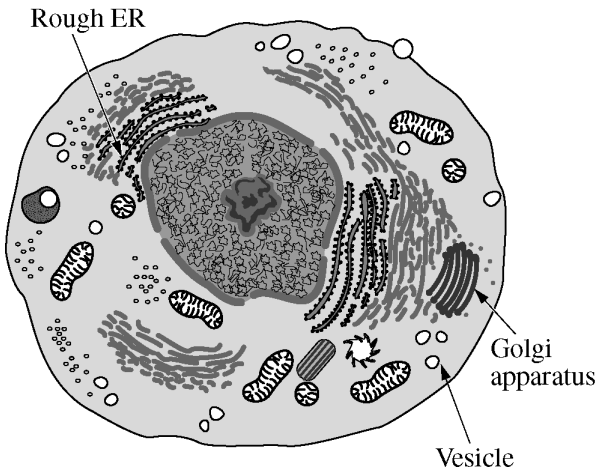
(A) Structure *Y* contributes the raw materials required for the synthesis of proteins.

(B) Structure *Y* packages proteins for export from the cell.

(C) Structure *Y* is the location where proteins are synthesized. ✓

(D) Structure *Y* contains enzymes that cut and activate proteins.

115. The figure below illustrates a eukaryotic cell. Which of the following best describes how the three structures indicated by the arrows work together?



(A) To synthesize lipids and modify toxic substances in order to render them harmless

(B) To synthesize and isolate proteins for secretion or for use in the cell ✓

(C) To catabolize nutrients and produce ATP for intracellular energy storage

(D) To synthesize all ribosomal proteins

2 Cell MCQ

116. A certain type of specialized cell contains an unusually large amount of rough endoplasmic reticulum (ER).

Which of the following functions is this cell type most likely specialized to perform?

(A) The production and secretion of steroids

(B) The destruction of toxic materials produced in other cells of the organism

(C) The synthesis of polysaccharides for energy storage

(D) The production and secretion of proteins ✓

Answer D

Correct. The cytosolic surface of the rough ER is covered by ribosomes that synthesize proteins that are then transported into the rough ER, then to the Golgi complex, and finally out of the cell.

117. A scientist is studying the various prokaryotic and eukaryotic species found floating in a sample of water taken from a marine ecosystem.

Which cellular component will be found in the widest range of organisms in the sample?

(A) The chloroplast, since all organisms need a source of energy.

(B) The ribosome, since all organisms need to synthesize proteins. ✓

(C) The mitochondrion, since all organisms need to break down glucose.

(D) The cell wall, since all marine organisms need them for support.

Answer B

Correct. Ribosomes are found in all forms of life, allowing for comparison of the widest possible range of plankton species.

2 Cell MCQ

118. In an experiment, researchers provided a radiolabeled amino acid to living plant cells. After one hour, the researchers determined the amount of the radiolabeled amino acid that was in each of several subcellular compartments. The results of the experiment are represented in the table.

RELATIVE AMOUNTS OF RADIOLABELED AMINO ACID

Nucleus	Mitochondria	Endoplasmic Reticulum	Cytosol
2.1	2.7	1.9	1

Which of the following conclusions about the radiolabeled amino acid is best supported by the results of the experiment?

(A) It was mostly incorporated into nucleic acids that store the biological information.

(B) It was mostly incorporated into proteins that regulate and manage metabolic reactions. ✓

(C) It was mostly incorporated into lipids that help separate cells from their surrounding environment.

(D) It was mostly incorporated into carbohydrates that form protective structures outside the cells.

Answer B

Correct. Amino acids are the building block of proteins, and the data indicate that most of the radiolabeled amino acid was in the mitochondria. Therefore, the data best support the conclusion that most of the amino acid molecules were incorporated into proteins that regulate and manage the metabolic reactions that occur in mitochondria.

2 Cell MCQ

119. Researchers have proposed a model of the process by which a newly synthesized protein is transported to the plasma membrane and secreted into the extracellular space. The model is represented in Figure 1.

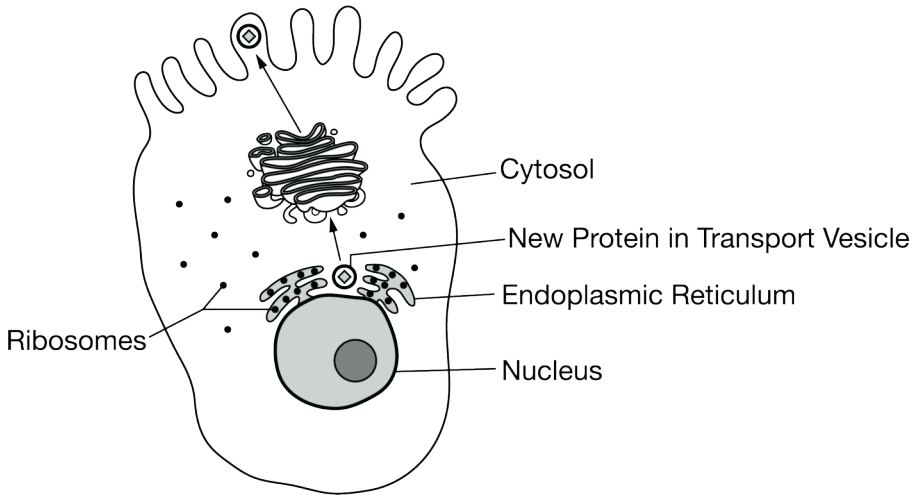


Figure 1. A model of the intracellular transport of a newly synthesized secreted protein

Based on the model, the newly synthesized protein is transported directly from the endoplasmic reticulum to which of the following?

(A) The nucleus

(B) The plasma membrane

(C) The Golgi complex ✓

(D) The extracellular space

Answer C

Correct. Membrane proteins and secreted proteins are synthesized by ribosomes bound to the surface of the endoplasmic reticulum. As shown in the figure, a newly synthesized secreted protein is transported by vesicles from the endoplasmic reticulum directly to the Golgi complex.

120. Macrophages and B lymphocytes are cells that help fight infections. Macrophages engulf and destroy bacteria. B lymphocytes produce proteins that bind to bacteria in the blood.

Based on the functions of macrophages and B lymphocytes, which of the following claims is most likely true of the organelles needed in both types of cells?

2 Cell MCQ

(A) They need organelles that contain enzymes for digesting materials.

(B) They need organelles that store macromolecules and water.

(C) They need organelles that process and transport proteins. ✓

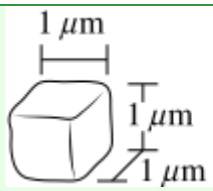
(D) They need organelles that produce lipids for the plasma membrane.

Answer C

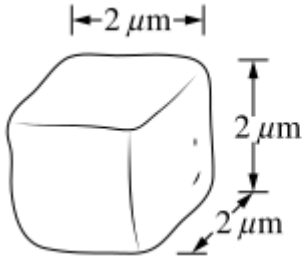
Correct. Both cell types require an extensive endoplasmic reticulum and Golgi complex to process and transport proteins. Macrophages need these organelles for producing digestive enzymes, while B lymphocytes need them for protein (antibody) production and secretion.

121. Which one of the following cells is likely to be the most efficient at eliminating waste by diffusion?

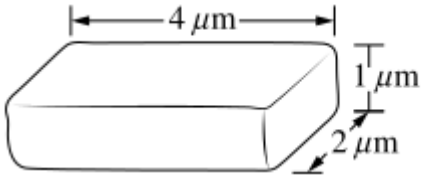
(A) ✓



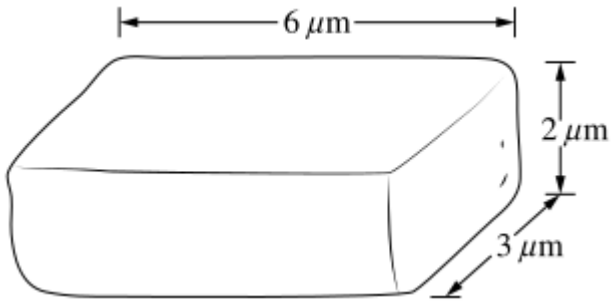
(B)



(C)



(D)



2 Cell MCQ

Answer A

Correct. In cells, the elimination of wastes by diffusion is limited by the ratio of surface area to volume. The greater the ratio of surface area to volume is, the more efficient the diffusion of materials from the cell is. This cell has the greatest surface-area-to-volume ratio ($6:1$), making it the most efficient at eliminating waste.

122. Which of the following best describes the relationship between the size and shape of a cell and the ability of the cell to move substances across the membrane?

(A) Larger and more round-shaped cells are more efficient at moving substances across the membrane than small round-shaped cells.

(B) Smaller and more oval-shaped cells release waste more slowly than large oval-shaped cells.

(C) Larger cube-shaped cells obtain nutrients more quickly than small cube-shaped cells.

(D) Smaller irregularly shaped cells are more efficient at dissipating heat than large irregularly shaped cells. ✓

123. **TABLE 1. DIMENSIONS OF SHAPES USED TO MODEL EFFICIENT MATERIAL EXCHANGE**

Cell Shape	Width at Widest Point (μm)	Height (μm)	Surface Area (μm^2)	Volume (μm^3)
Right circular cylinder	30	90	9,896	63,617
Triangular prism	30	90	8,879	35,074
Rectangular prism	30	90	12,600	81,000
Square-based pyramid	30	90	6,374	27,000

A scientist is developing a mathematical model of cells of different shapes. To construct the model, the scientist has specified that the width of each cell at its widest point must be $30\ \mu m$ and the height of each cell must be $90\ \mu m$. Table 1 shows the three-dimensional shapes that the scientist is considering for the model cells. Which of the proposed shapes for the model cells will allow the most efficient exchange of materials with the surrounding environment?

(A) Right circular cylinder

(B) Triangular prism ✓

(C) Rectangular prism

(D) Square-based pyramid

2 Cell MCQ

Answer B

Correct. This cell shape has the greatest surface area-to-volume ratio (0.253), allowing for the most efficient exchange of materials with the surrounding environment.

124.

Cellular Shape	Ratio of Surface Area to Volume
Cube	1.7
Regular Tetrahedron	0.5
Cylinder	0.9
Sphere	0.6

In an experiment, the efficiency of oxygen exchange across the plasma membrane is being assessed in four artificial red blood cells. The table above lists some properties of those artificial cells. Other conditions being equal, which artificial cell is predicted to be the most efficient in exchanging oxygen with the environment by diffusion?

(A) The cuboidal cell ✓

(B) The tetrahedral cell

(C) The cylindrical cell

(D) The spherical cell

125. Some cells, such as intestinal cells, exchange a lot of material with their surroundings. The surface-to-volume ratio of these cells affects the efficiency of material exchange.

The table provides measurements of four different eukaryotic cells.

Cell	1	2	3	4
Total surface area (μm^2)	40	60	80	100
Total volume (μm^3)	20	10	30	20

Based on the data, which cell is likely to be most effective in the exchange of materials?

(A) Cell 1

(B) Cell 2 ✓

(C) Cell 3

(D) Cell 4

2 Cell MCQ

Answer B

Correct. The surface area to volume calculation is $\frac{60}{10} = 6$. Of the four cells, this one has the highest ratio of surface area to volume and is likely to be most effective in the exchange of materials.

126.

TABLE 1. AVERAGE DIMENSIONS OF FOUR CELL TYPES

	Surface Area (mm^2)	**Volume** (mm^3)
Cell type 1	96	64
Cell type 2	60	10
Cell type 3	54	27
Cell type 4	24	8

Which of the cell types represented in Table 1 is most efficient at acquiring nutrients by diffusion?

(A) Cell type 1

(B) Cell type 2 ✓

(C) Cell type 3

(D) Cell type 4

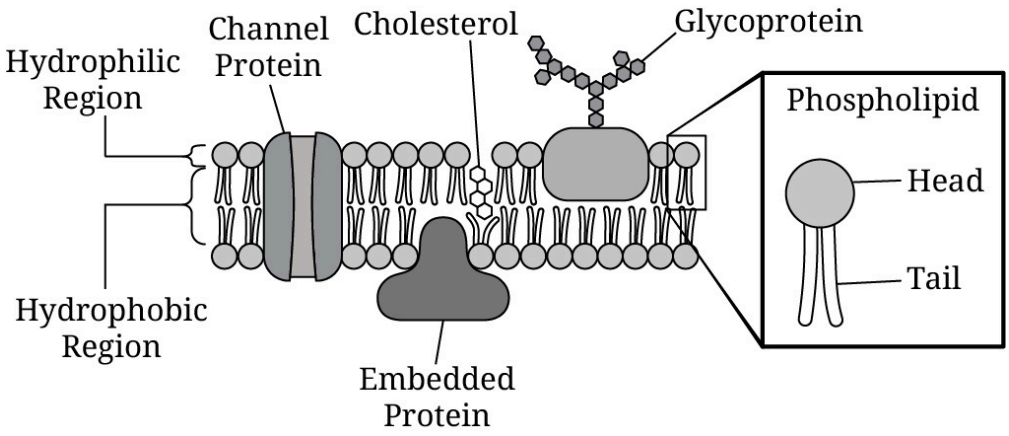
Answer B

Correct. Nutrient acquisition by diffusion relies on surface area to volume ratio. $\frac{60}{10} = 6$. Cell type 2 has the largest SA /V ratio and would therefore be the most efficient at acquiring nutrients by diffusion.

2 Cell MCQ

127. Figure 1 shows a cell membrane that is composed of a phospholipid bilayer and several other components.

Figure 1. Components of a cell membrane



Based on Figure 1, which of the following membrane components are most likely nonpolar?

(A) The phospholipid tails and cholesterol ✓

(B) The phospholipid heads and the channel protein

(C) The phospholipid heads and the glycoprotein

(D) The phospholipid tails and the embedded protein

128. Which of the following best describes the position of the phospholipids within the phospholipid bilayer of the plasma membrane?

(A) The hydrophilic heads face the inner region of the membrane, and the hydrophobic tails point outward.

(B) The hydrophilic tails face the inner region of the membrane, and the hydrophobic heads point outward.

(C) The hydrophobic heads face the inner region of the membrane, and the hydrophilic tails point outward.

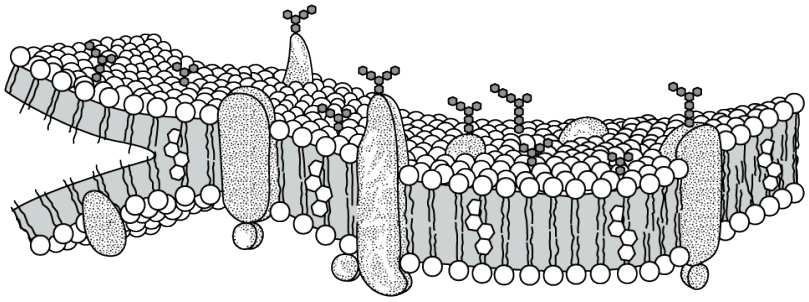
(D) The hydrophobic tails face the inner region of the membrane, and the hydrophilic heads point outward. ✓

Answer D

Correct. Because both the extracellular fluid and the cytosol are aqueous, the hydrophilic heads of the phospholipids face these two fluids, while the hydrophobic tails face away from them toward the interior of the membrane.

2 Cell MCQ

129. The model below shows the structure of a portion of a plasma membrane in an animal cell.



Which statement best explains the orientation of the phospholipid molecules in this model?

(A) The nonpolar portions of the phospholipid molecules are attracted to the internal and external environments.

(B) The hydrophilic phosphate groups of the phospholipid molecules are attracted to the aqueous internal and external environments. ✓

(C) The embedded proteins attract the fatty acid tails of the phospholipids, so the tails point away from the internal and external aqueous environments.

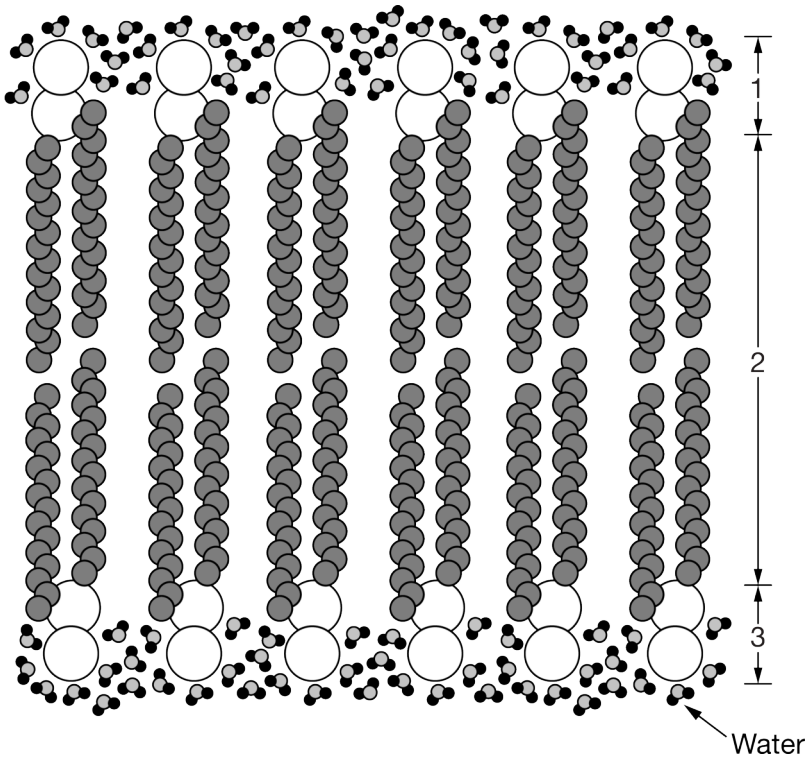
(D) The fatty acid tails of the phospholipid molecules are hydrophilic and are repelled by the internal and external aqueous environments.

Answer B

Correct. The hydrophilic polar phosphate groups of the phospholipid molecules orient toward the polar internal and external aqueous environments, forming a bilayer that keeps the hydrophobic portions of the phospholipids away from the polar molecules.

2 Cell MCQ

130.



Which of the following best describes the numbered areas?

(A) Areas 1 and 3 are polar, since the membrane molecules are aligned with water molecules. ✓

(B) Area 2 is polar, since water has been excluded from this area of the membrane.

(C) Areas 1 and 3 are hydrophilic, since membrane molecules formed covalent bonds with water.

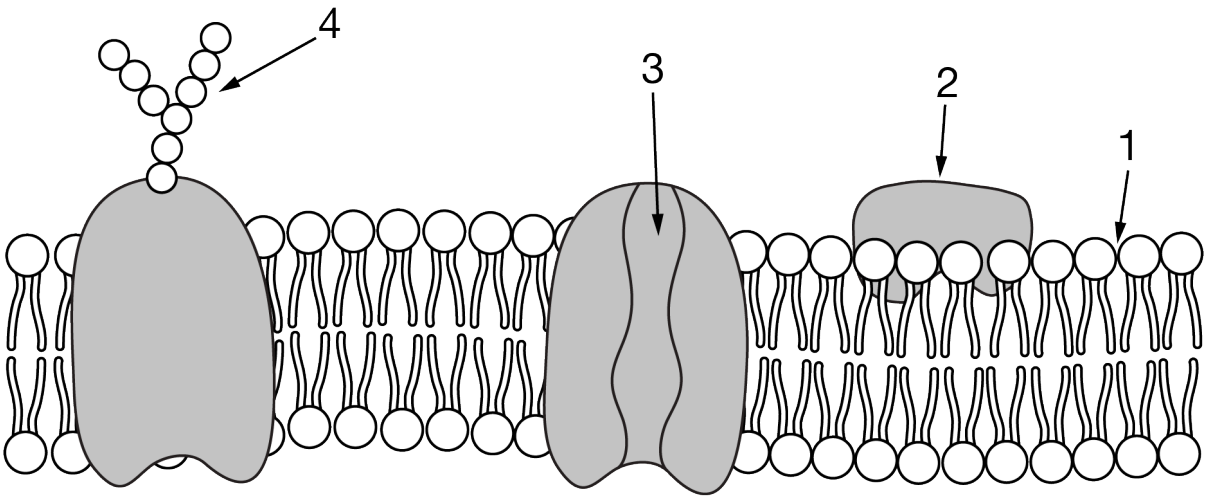
(D) Area 2 is nonpolar, since hydrogen bonds between the adjacent lipids hold the membrane together.

Answer A

Correct. As indicated in the diagram, the heads of the phospholipids in areas 1 and 3 are adjacent to water molecules, which shows that the phospholipid heads are hydrophilic.

2 Cell MCQ

131. Figure 1. Testosterone movement across the cellular membrane



Testosterone is a small steroid hormone that is important in cell signaling. Which of the following describes where testosterone enters a cell and why it is able to cross at that point?

(A) 1, testosterone is nonpolar and can diffuse through the membrane. ✓

(B) 2, testosterone covalently binds to a surface protein and transports into the cell.

(C) 3, testosterone dissolves in water and flows through the channel.

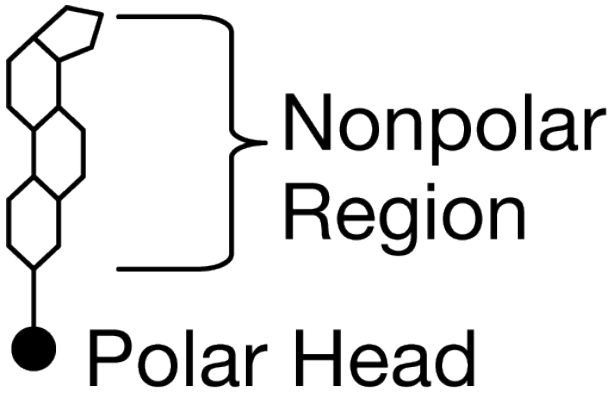
(D) 4, testosterone is filtered out of the extracellular fluid and taken into the cell by endocytosis.

Answer A

Correct. Steroids such as testosterone are hydrophobic lipids. Therefore, testosterone can cross the hydrophobic inner region of the phospholipid bilayer.

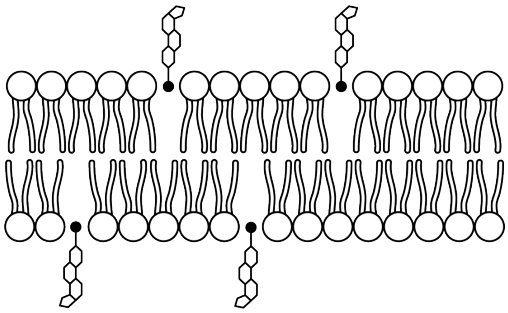
2 Cell MCQ

132. Cholesterol is a naturally occurring substance that helps regulate the fluidity of a cell's plasma membrane. A cholesterol molecule can be represented as having a polar head and a nonpolar region, as shown in the figure.

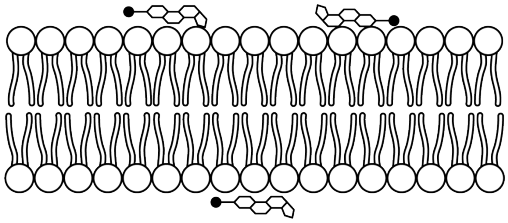


Which of the following models shows how cholesterol molecules most likely interact with the phospholipid bilayer of a cell's plasma membrane?

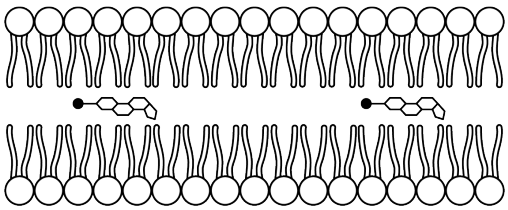
(A)



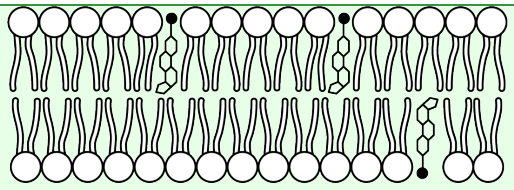
(B)



(C)



(D) ✓



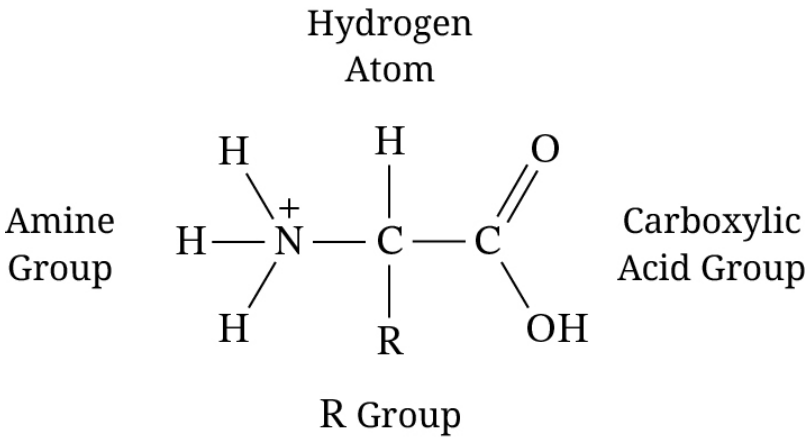
2 Cell MCQ

Answer D

Correct. The model correctly shows the polar heads of the cholesterol molecules interacting with the polar heads of the phospholipids. Also, the model correctly shows the nonpolar regions of the cholesterol molecules interacting with the hydrophobic interior of the phospholipid bilayer.

133. Figure 1 shows the general structure of an amino acid.

Figure 1. Structure of an amino acid



Which of the following amino acid components shown in Figure 1 best determines the interaction between an embedded protein and the hydrophobic regions of the cell membrane?

(A) The hydrogen atom bonded to the central carbon

(B) The carboxyl group bonded to the central carbon

(C) The R group bonded to the central carbon ✓

(D) The amine group bonded to the central carbon

134. A spherical bacterial cell has a radius of $3\ \mu\text{m}$. The human egg cell has a radius of $100\ \mu\text{m}$.

Which statement correctly indicates the cell that is able to more efficiently exchange materials with the external environment and provides a correct explanation?

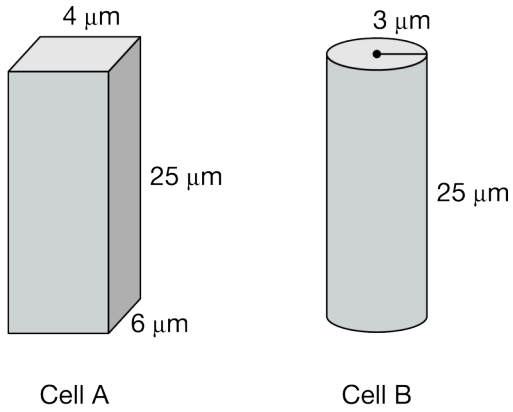
2 Cell MCQ

(A) The egg cell, because it has the smallest surface-to-volume ratio.
(B) The egg cell, because it has the largest surface-to-volume ratio.
(C) The bacterial cell, because it has the smallest surface-to-volume ratio.
(D) The bacterial cell, because it has the largest surface-to-volume ratio. ✓

Answer D

Correct. The bacterial cell is more efficient in the exchange of materials with the external environment. While the egg cell is much larger than the bacterial cell, the egg cell has a surface-to-volume ratio that is 33 times smaller than the bacterial cell's surface-to-volume ratio.

135. Two different models of a living cell are represented in the figure.



Note: Figures not drawn to scale.

Of the two cells represented in the figure, which would likely be more efficient at exchanging substances with the surrounding environment?

(A) Cell A, because it has the larger surface-area-to-volume ratio. ✓
(B) Cell A, because it has the smaller surface-area-to-volume ratio.
(C) Cell B, because it has the larger surface-area-to-volume ratio.
(D) Cell B, because it has the smaller surface-area-to-volume ratio.

Answer A

Correct. Of the two cells represented in the figure, Cell A has the larger surface-area-to-volume ratio of

2 Cell MCQ

0.91 compared to a surface-area-to-volume ratio of 0.75 for Cell B. Cell A would likely exchange substances with the surrounding environment at a greater rate.

136. Stomata are pores on the surfaces of the leaves and stems of plants that regulate gas exchange between the plants and the atmosphere.

Researchers found that the stomata density on the leaves of a species of plant change as the concentration of CO_2 in the atmosphere changes. When grown at 350 $\text{ppm } CO_2$ the plant has an average density of 300 stomata per mm^2, but when grown at 400 $\text{ppm } CO_2$ the plant has an average density of 250 stomata per mm^2.

Which of the following best describes how the ratio of the density of stomata (stomata per mm^2) per CO_2 concentration ($\text{ppm } CO_2$) changes as the CO_2 concentration increases?

(A) The ratio decreases from 0.86 to 0.63, because fewer stomata are needed at higher CO_2 concentrations. ✓

(B) The ratio decreases from 1.6 to 1.2, because fewer stomata are needed at higher CO_2 concentrations.

(C) The ratio increases from 0.63 to 0.86, because more stomata are needed at higher CO_2 concentrations.

(D) The ratio increases from 1.2 to 1.6, because more stomata are needed at higher CO_2 concentrations.

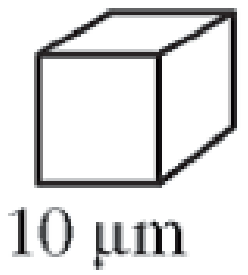
Answer A

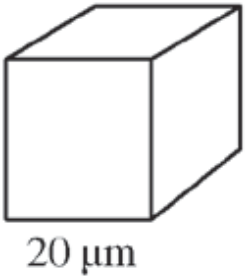
Correct. The ratio of 300 stomata per mm^2 to 350 $\text{ppm } CO_2$ is 0.86, and the ratio of 250 stomata per mm^2 to 400 $\text{ppm } CO_2$ is 0.63. This reflects that fewer stomata are needed as the concentration of CO_2 increases.

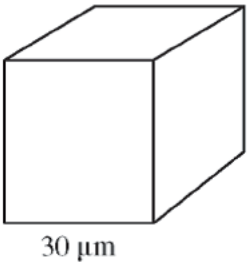
137. Simple cuboidal epithelial cells line the ducts of certain human exocrine glands. Various materials are transported into or out of the cells by diffusion. (The formula for the surface area of a cube is 6 X S^2, and the formula for the volume of a cube is S^3, where S = the length of a side of the cube.)

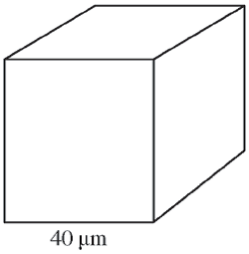
Which of the following cube-shaped cells would be most efficient in removing waste by diffusion?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates the ability to calculate surface area-to- volume ratios and predict their effects on a cell's ability to exchange materials with its environment. The cell shown in A has the greatest surface area-to- volume ratio at 0.6:1.

2 Cell MCQ

138. Plant cell walls are composed of cellulose, while fungal cell walls are composed of chitin. A group of scientists hypothesize that this difference means the cell wall has largely different functions in plant cells and fungal cells. Alternatively, another group of scientists hypothesize that despite their biochemical differences, plant and fungal cell walls serve similar functions.

Which of the following observations would best support the alternative hypothesis described above?

(A) Plant cell walls are found just outside the plasma membrane, while fungal cell walls are found just beneath the plasma membrane.

(B) In both plant cells and fungal cells, the cell wall surrounds the outside of the cell membrane. ✓

(C) Some plant cells have secondary cell walls that confer additional rigidity, while fungal cells do not.

(D) Photosynthesis occurs in plant cells, but it does not occur in fungal cells.

Answer B

Correct. In both plants and fungi, a major function of the cell wall is to confer structural rigidity to the cell. Thus, in both cell types, the cell wall's location is similar. This is evidence supporting the alternative hypothesis proposed above.

139. Which of the following best describes the primary mechanism of transport whereby carbon dioxide moves from inside a cell to outside the cell?

(A) Carbon dioxide is released when vesicles containing carbon dioxide fuse with the plasma membrane.

(B) Carbon dioxide is actively pumped into the membrane between the halves of the bilayer and then passively diffuses out.

(C) Carbon dioxide passively diffuses between the membrane lipids. ✓

(D) Carbon dioxide passively moves through membrane channel proteins.

Answer C

Correct. Because CO_2 is a nonpolar gas, it moves directly between the cell membrane's hydrophobic lipid molecules, from an area of high concentration to an area of lower concentration.

2 Cell MCQ

140. Beetroot cells contain a family of dark red pigments called betalains. The selectively permeable nature of the beetroot cells keeps the internal environment of the cell separate from the external environment of the cell. Researchers are interested in determining whether the selective permeability of beetroot cells is due to the cell membrane or if it is due to the cell wall.

Exposure to cellulase is known to damage the structure of the cell wall. An experiment is set up in which beetroot cells are placed in an aqueous solution with cellulase and in one without cellulase.

Which of the following results best refutes the alternative hypothesis that selective permeability is a consequence of the cell wall?

(A) When beetroot cells are placed in a solution with cellulase, the solution turns dark red.

(B) When beetroot cells are placed in a solution with cellulase, the solution remains clear. ✓

(C) When beetroot cells are placed in a solution, it turns dark red with or without cellulase present.

(D) Since plant cells contain cell membranes, not cell walls, the alternate hypothesis cannot be tested.

Answer B

Correct. Cellulase digests cellulose and damages the structure of the cell wall, not the cell membrane. The lack of color change in the solution indicates that the betalain is not leaking out of the beetroot cells even though the cell wall has been damaged. This refutes the alternative hypothesis.

2 Cell MCQ

141. A model of the plasma membrane showing several biological molecules, including a transmembrane protein, is shown in Figure 1.

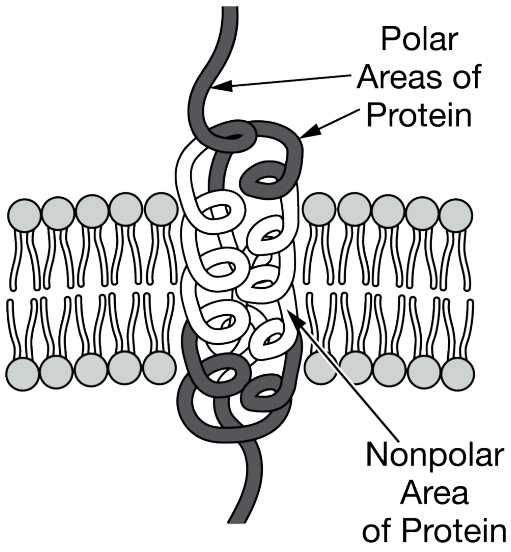


Figure 1. Phospholipid bilayer with transmembrane protein

Which statement best explains why correct protein folding is critical in the transmembrane protein shown above?

(A) Interactions of the hydrophobic and hydrophilic amino acids help to anchor the protein in the membrane. ✓

(B) Interactions of the peptide bonds of the protein with the membrane will affect the rate at which substances can cross the membrane.

(C) Interactions of the protein and phospholipids increase membrane fluidity.

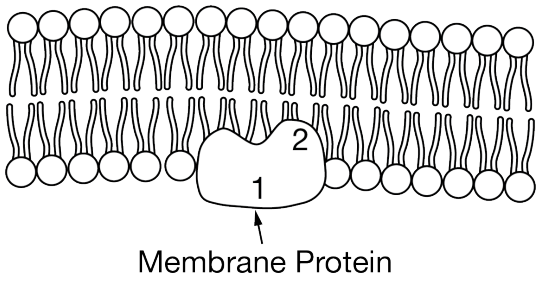
(D) Interactions of the quaternary structure of the protein will increase hydrogen bonding in the membrane and make the membrane less fluid.

Answer A

Correct. Tertiary structure of the protein resulting from hydrophobic and hydrophilic interactions plays a key role in how the protein interacts with the hydrophobic tails and hydrophilic head of the membrane phospholipids.

2 Cell MCQ

142. The figure shows a representation of a protein embedded in a cell membrane. The numbers indicate different structural regions of the protein.



Based on the figure, which of the following statements best describes the relationship between regions 1 and 2 of the protein?

(A) Region 1 is hydrophilic because it interacts with the interior of the membrane, whereas region 2 is hydrophobic because it interacts with an aqueous environment.

(B) Region 1 is hydrophilic because it interacts with an aqueous environment, whereas region 2 is hydrophobic because it interacts with the interior of the membrane. ✓

(C) Region 1 is hydrophobic because it interacts with the interior of the membrane, whereas region 2 is hydrophilic because it interacts with an aqueous environment.

(D) Region 1 is hydrophobic because it interacts with an aqueous environment, whereas region 2 is hydrophilic because it interacts with the interior of the membrane.

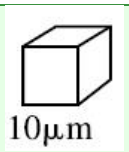
Answer B

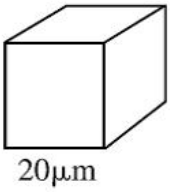
Correct. A cell membrane is a phospholipid bilayer that separates one aqueous environment from another. The interior of a phospholipid bilayer is a hydrophobic environment. Because region 1 interacts with the aqueous environment on one side of the phospholipid bilayer, it is most likely hydrophilic. Because region 2 interacts with the interior of the phospholipid bilayer, it is most likely hydrophobic.

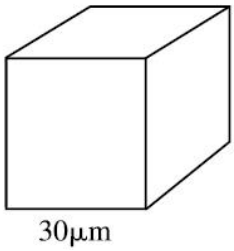
143. Simple cuboidal epithelial cells line the ducts of certain human exocrine glands. Various materials are transported into or out of the cells by diffusion. (The formula for the surface area of a cube is 6 X S^2, and the formula for the volume of a cube is S^3, where S = the length of a side of the cube.)

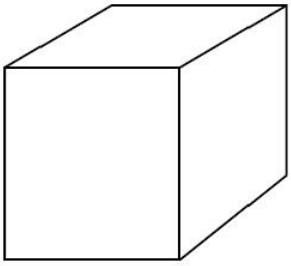
Which of the following cube-shaped cells would be most efficient in removing waste by diffusion?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

144. Researchers propose a model to explain variation in phytoplankton cell sizes in a marine environment. They base their model on the idea that smaller cells absorb nutrients more efficiently. The researchers predict that the mean diameter of phytoplankton cells will change by 50 micrometers for every 5-kilometer increase in distance from the shore because of a gradual decrease in nutrient availability. To test their model, the researchers determine that the phytoplankton cells found closest to shore have a mean diameter of 900 micrometers.

Based on the model, what will be the mean diameter of the phytoplankton cells that are found 25 kilometers from shore?

(A) 650 micrometers

(B) 875 micrometers

(C) 925 micrometers

(D) 1150 micrometers

2 Cell MCQ

Answer A

Correct. The model predicts that phytoplankton cell size will decrease with increasing distance from the shore, resulting in increased efficiency of nutrient absorption. The correct value was arrived at by subtracting 250 from 900.

145. The table below provides a comparison of nitrogenous waste production in selected organisms.

Nitrogenous Waste	Solubility in Water	Amount of Water Required for Excretion	Organisms that Primarily Produce the Nitrogenous Waste
Ammonia	High	High	Freshwater fish, aquatic invertebrates
Urea	Medium	Medium	Mammals, sharks
Uric acid	Very low	Very low	Birds, reptiles, and most terrestrial insects and arthropods

Which of the following statements is most consistent with the data in the table?

(A) In response to the hypotonic environment in which freshwater fish live, they excrete ammonia in concentrated urine or across their gills.

(B) The kidneys of reptiles and birds are highly efficient because little water is needed to excrete uric acid. ✓

(C) Birds excrete ammonia in addition to uric acid, and the ratio of the two substances is independent of whether the birds are primarily terrestrial or aquatic species.

(D) The similar regulation of extracellular fluid volume and composition in all the organisms suggests conservation of kidney structure throughout evolution.

146. Hereditary spherocytosis (HS) is a disorder of red blood cells that causes the cells to be smaller and spherical instead of having the usual flattened, biconcave shape. The average diameter of normal red blood cells is $7.2\ \mu\mathrm{m}$, and the average diameter of red blood cells in a person with HS was found to be $6.7\ \mu\mathrm{m}$. The normal red blood cell has an average surface area of $136\ \mu\mathrm{m}^2$ and an average volume of $91\ \mu\mathrm{m}^3$.

Which of the following provides an accurate calculation of the surface area to volume ratio of an HS red blood cell, as well as a prediction of its effect on the efficient transferring of oxygen compared to a normal red blood cell?

(A) The ratio is 0.45, and the cells are more efficient at transferring oxygen.

(B) The ratio is 1.12, and the cells are less efficient at transferring oxygen.

(C) The ratio is 0.89, and the cells are less efficient at transferring oxygen. ✓

(D) The ratio is 141, and the cells are more efficient at transferring oxygen.

2 Cell MCQ

Answer C

Correct. The correct calculation of the surface area to volume ratio of the HS cell is 0.89. This ratio is less than the ratio found in a normal red blood cell, 1.49, so the cell would be less efficient at transferring oxygen.

EFFECT OF WATER-SOLUBLE POLLUTANTS ON MEMBRANE PERMEABILITY

Treatment Group	Treatment Solution*	Mean Absorbance of 460 nm Light ($n = 5$)	2×Standard Error of the Mean (2×SEM)
I	70% isopropanol; 30% water	0.164	0.032
II	90% isopropanol; 10% water	0.125	0.100
III	50% acetone; 50% water	0.215	0.034
IV	70% acetone; 30% water	0.274	0.018
V	100% water	0.095	0.004

*Concentrations are reported as percent by volume.

A student formulated a hypothesis that water-soluble pollutants damage living organisms by increasing the permeability of cellular membranes. To test the hypothesis, the student investigated the effect of isopropanol and acetone on beet root cells. The vacuoles of beet root cells contain large amounts of betacyanin, a water-soluble pigment that is released into the extracellular environment as a result of increased membrane permeability.

The student prepared identical samples of beet root tissue and incubated each sample for 15 minutes in the specific solution for that group. At the end of the incubation period, the student measured the absorbance of 460 nm light for each sample. A greater concentration of betacyanin in the solution surrounding the beet root cells results in a greater absorbance of 460 nm light. The results of the experiment are shown in the table above.

147. Based on the data from the investigation, which of the following is the best scientific question about organisms living in water that is polluted with organic solvents?

(A) Do organisms without pigments have a selective advantage in polluted environments?

(B) Will organisms living in polluted environments exhibit detrimental effects from an increased permeability of their cellular membranes? ✓

(C) Will organic solvents from the environment become incorporated into the cell membranes of organisms living in polluted environments?

(D) Should governments place more stringent restrictions on the release of organic solvents into aquatic ecosystems?

148. The student analyzed the data from the investigation and concluded that the estimate of the mean of one treatment group was unreliable. Which of the following identifies the treatment group most likely to have provided an unreliable estimate of the mean, and correctly explains why the estimate appears unreliable?

2 Cell MCQ

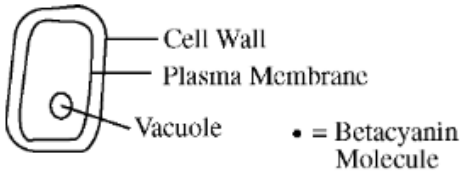
(A) Treatment group II; it has a lower than expected mean absorbance and the largest standard error of the mean. ✓

(B) Treatment group III; it has a higher than expected mean absorbance and the largest standard error of the mean.

(C) Treatment group IV; it has a higher than expected mean absorbance and the smallest standard error of the mean.

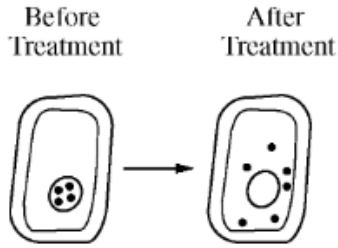
(D) Treatment group V; it has a lower than expected mean absorbance and the smallest standard error of the mean.

149.

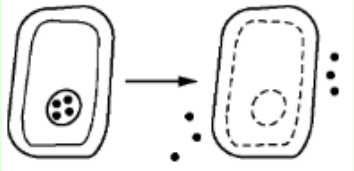


The illustration above is a model of a typical beet root cell. Based on the experimental results, which of the following best represents the effect of acetone on the permeability of cellular membranes?

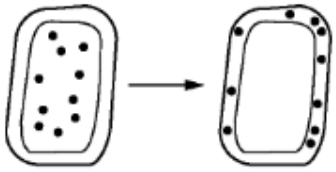
(A)



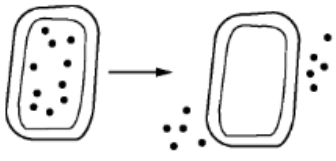
(B) ✓



(C)



(D)



150. Which of the following is the dependent variable in the experiment?

2 Cell MCQ

(A) The percent by volume of isopropanol in the treatment solutions

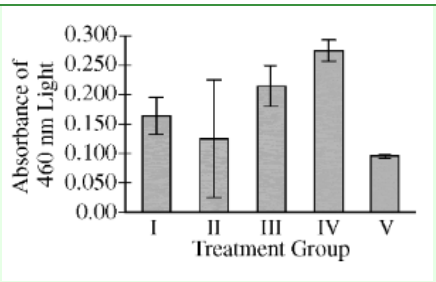
(B) The percent by volume of water in the treatment solutions

(C) The length of time each sample was incubated in the treatment solutions

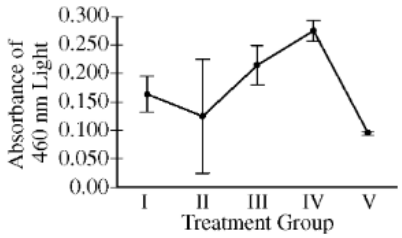
(D) The absorbance of 460 nm light by the treatment solutions ✓

151. Which of the following graphs is the most appropriate representation of the experimental data?

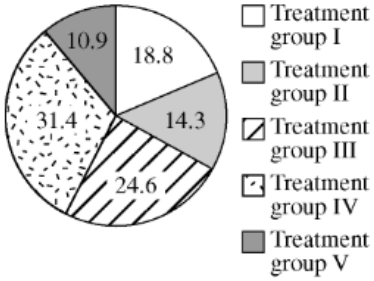
(A) ✓



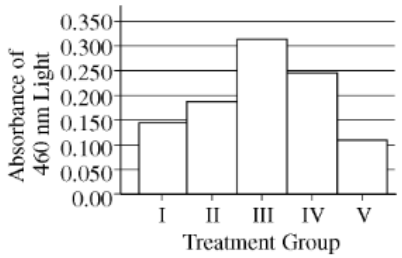
(B)



(C)



(D)



2 Cell MCQ

152. Aldosterone (a steroid hormone) is a small, nonpolar, hydrophobic molecule that enters a target cell by moving across the plasma membrane, down a concentration gradient.

Based on the information presented, how does aldosterone most likely enter target cells?

(A) By simple diffusion because aldersterone can enter a cell by moving across the plasma membrane and moving down the concentration gradient. ✓

(B) By facilitated diffusion because aldersterone would need the assistance of a protein channel to move across the plasma membrane.

(C) By active transport because aldersterone would need energy to move across the plasma membrane.

(D) By endocytosis because aldersterone would need to be transported by vesicle across the plasma membrane

Answer A

Correct. Small, nonpolar, hydrophobic molecules can enter a cell by moving across the plasma membrane and down a concentration gradient by simple diffusion.

153. Two competing hypotheses exist regarding the cell membrane structure. One hypothesis states that membrane structure is static and membrane components throughout the bilayer are rigidly bound. Alternatively, the other hypothesis states that cell membranes are a fluid mosaic in which membrane components may drift within the bilayer around the surface of the cell. An experiment is set up in which membrane proteins of two different cells are fluorescently labeled with two different colors and then fused as shown in Figure 1.

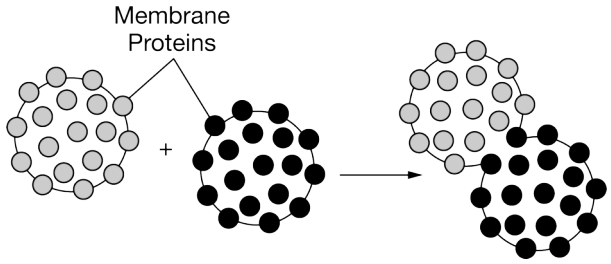
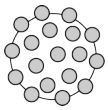
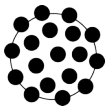


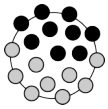
Figure 1. Model of initiation of cell fusion experiment

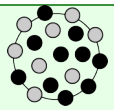
Which of the following results, one hour after membrane fusion, best supports the alternative hypothesis that the cell membrane is a fluid mosaic?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

Answer D

Correct. The membrane proteins from each cell have mixed and drifted within the bilayer, which supports the fluid mosaic model of cell membranes. Cell membranes consist of a structural framework of phospholipid molecules that is embedded with proteins and steroids that may flow around the surface of the cell within the bilayer.

154. Researchers exposed identical cells to solutions containing different solutes. Each solution had a higher concentration of the solute outside the cells than inside the cells. The solutes varied in size, charge, and polarity. After a set time, the researchers measured the concentration of each solute inside the cells to determine the permeability of the cell membranes to each solute. The results are shown in Table 1.

Table 1. Permeability of Cell Membranes to Different Solutes

Solute	Size	Charge	Polarity	Initial Concentration outside the cell	Concentration Inside the Cell
A	Small	Neutral	Nonpolar	High	High
B	Large	Positive	Nonpolar	High	Low
C	Small	Negative	Polar	High	Medium

Based on the data in Table 1, which of the following statements best supports the hypothesis that only certain types of molecules can move freely across the cell membranes?

2 Cell MCQ

(A) Large, nonpolar solutes showed an increase in the concentration inside the cell, while small, nonpolar solutes showed a slow increase in the concentration inside the cell.

(B) Ionic solutes showed equal movement of solutes inside and outside the cell internal and external concentrations.

(C) Small, nonpolar solutes showed the greatest increase in internal concentration, while large, nonpolar solutes showed little to no increase. ✓

(D) None of the solutes showed any change in internal concentration, regardless of their size, polarity, and charge.

155. Which of the following statements is true regarding the movement of substances across cell membranes?

(A) Ions are unable to move through the phospholipid bilayer because the nonpolar tail regions of the phospholipids are hydrophobic. ✓

(B) Ions are able to move through the phospholipid bilayer because the polar head regions of the phospholipids are charged.

(C) Water is able to move through the phospholipid bilayer because the nonpolar tail regions of the phospholipids are charged.

(D) Water is unable to move through the phospholipid bilayer because the polar head regions of the phospholipids are charged.

156. A team of biologists develop a new drug, and one team member hypothesizes that the drug is incapable of freely passing across the plasma membrane and requires the help of membrane proteins to enter cells. Alternatively, another biologist on the team hypothesizes that the drug can diffuse passively across the plasma membrane like O_2 and CO_2 can.

Which of the following, if true about the drug, best supports the alternative hypothesis that the new drug will exhibit simple diffusion across plasma membranes?

(A) The drug is a small nonpolar molecule. ✓

(B) The drug is a small charged molecule.

(C) The drug is a large polar molecule.

(D) The drug is a large charged molecule.

Answer A

Correct. Small nonpolar molecules freely pass across the plasma membrane. If the drug is a small nonpolar molecule, this supports the alternative hypothesis.

2 Cell MCQ

157. The cell membrane is selectively permeable due to its structure. Thus, the internal environment of the cell is distinct from the external environment of the cell. One biologist hypothesizes that small nonpolar molecules readily pass through the membrane. Another biologist alternatively hypothesizes that these types of molecules require channel and transport proteins that are embedded in the membrane in order to move across the membrane.

Which of the following data would best refute this alternative hypothesis?

(A) Ethanol is found in the cytosol of cells when they are briefly exposed to a ten percent ethanol solution.

(B) Cells become oxygen deficient when membrane protein activity is blocked.

(C) CO_2 and N_2 movement in and out of cells is unaffected when membrane protein activity is blocked. ✓

(D) Sodium ions cannot move across the cell membrane when membrane protein activity is blocked.

Answer C

Correct. Both CO_2 and N_2 are small nonpolar molecules. Their ability to enter and leave cells normally when membrane protein activity is blocked would refute the alternative hypothesis.